Absolute Convergence and Taylor Expansion in Web Based Models of Linear Logic

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Abstract

We provide in this paper a generic construction of web models of linear logic with partial sums. Our construction captures a wide class of orthogonality models, ranging from coherence spaces to probabilistic coherence spaces, finiteness spaces and Köthe spaces. All these models are built on the same principles, but were very heterogeneous in the specificities of their technical development. Our construction factorizes these specificities, allowing a unified treatment of further developments, such as differentiation and Taylor expansion.

The differential λ-calculus relates quantitative aspects of programs to differentiation and to Taylor expansion in models of linear logic. Recent work has generalized the axioms of Taylor expansion, so that they apply to many models that only feature partial sums. However, that work did not cover Köthe spaces and finiteness spaces. We generalize the theory of Taylor expansion to models in which coefficients can be negative, and we prove that our generic web models always satisfy this new axiomatization. Therefore, all the aforementioned models feature a Taylor expansion.

2012 ACM Subject Classification Theory of computation → Linear logic; Theory of computation → Categorical semantics; Mathematics of computing → Differential calculus

Keywords and phrases Categorical semantics, Linear Logic, Quantitative semantics, Taylor expansion

Digital Object Identifier 10.4230/LIPIcs.FSCD.2026.29

Related Version *Previous Version*: `https://hal.science/hal-05540426v1/file/hal.pdf`

Funding This work has been partly funded by the ANR JCJC project Species, Syntax and Semantics, ANR-20-CE48-0010.

Acknowledgements We would like to thank Thomas Ehrhard for his helpful advices. Our notion of absolute PCR was created following one of his suggestions. We also thank our reviewers for their numerous and helpful comments.

Introduction

Quantitative semantics was introduced with linear logic after Girard noticed that the interpretation of programs in mathematical models, were *analytic maps* [19]. They can be written as an infinite sum of multilinear maps through a formula called the Taylor expansion:

$$f(x) = \sum_{n\in\mathbb{N}} \frac{1}{n!} f^{(n)}(0)\cdot(x,\dots,x) \qquad \text{where } f^{(n)}(0) \text{ is the } n\text{-th derivative of } f \text{ at } 0.$$

The n-th derivative at 0 captures the part of the program that uses its input *exactly* n-times during execution. In this way, quantitative semantics helps us understand how computation interacts with non-determinism [33], probabilities [11, 15] or quantum primitives [41].

The differential λ-calculus [16] and the syntactic Taylor expansion [17] extend quantitative semantics. They give a purely syntactic way to track resources by building derivatives and Taylor expansion inside the language itself. These theories also uncover a strong link between the world of logic and computation, and the world of topology and analysis [12, 13, 26, 27].

The differential λ-calculus [6] and the syntactical Taylor expansion, and their models [37] have a fatal drawback: they allow any *countable* sums. The only way to interpret such sums computationally is as *non-determinism*. From a syntactic viewpoint, this is puzzling. Indeed,


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Taylor expansion can be applied to deterministic [17] or probabilistic [10] programs, and enjoys *uniformity* properties that are needed to prove normalization [17, 46]. From a semantic viewpoint, allowing countable sums rules out most quantitative semantics. Fortunately, recent works show that the differential calculus [14, 18] and the Taylor expansion [18] can be axiomatized using only *partial sums*. Partial sums match computational behaviors such as determinism in coherence spaces [19], or randomness in probabilistic coherence spaces [11].

One limitation remains. The infinite partial sums used in the *coherent* theory of Taylor expansion [18] rely on Σ-monoids [21]. Σ-monoids only allow positive sums: if $x+y=0$ then $x=y=0$. For programming language semantics, this is not a problem as programs never cancel each other out. However, this restriction becomes a serious issue when developing a theory of Taylor expansion that also works for quantitative semantics built on analysis and topology. In particular, Köthe spaces [12] and finiteness spaces [13] are important quantitative models that use analytic maps. Because they involve negative coefficients, they are not covered by the coherent Taylor expansion described in [18].

Contributions. We develop in Part I a generic construction of web models of linear logic with partial sums. Our construction unifies results on web models that were very similar in ideas, but featured very heterogeneous technical details. We first consider models in which objects are sets of vectors and morphisms are matrices whose coefficients range over an arbitrary Σ-monoid. Then, we generalize our construction to a more general notion of partial sums called PCMs [22]. We consider here PCMs whose summability is given by a notion of absolute convergence. This describes web models in which coefficients can be negative, as in Köthe spaces [12] and finiteness spaces [13].

In Part II, we generalize the coherent Taylor expansion of [18] to PCMs. This axiomatization of partial sum is more subtle to work with, in the same way that summability over real numbers is more subtle than summability over non-negative real numbers. We then prove in a unified way that *all* the web models developed in Part I feature such Taylor expansion.

Related work. The use of partial sums in semantics is rather common, and can be traced back to *algebraic programming* semantics [36]. Various notions of positive sums have been used in the literature, such as in geometry of interaction [21] and quantum programming semantics [43, 7]. Our results on web models provide a unified point of view on various proof techniques that appear throughout the literature on web based semantics [19, 12, 13, 11, 33]. Under some light additional assumptions (see Remark 32), our web models based on Σ-monoids can be described as categories of modules over Σ-semirings [45] with an orthogonal basis reminiscent of tight orthogonality in double glueing [23]. However, [45] does not describe Taylor expansion in that setting, and even though they prove that their model have a Lafont exponential, they do not provide any explicit definition. They also do not provide a definition for more general PCMs based on absolute convergence. Our theory of coherent Taylor expansion is an adaptation of the coherent Taylor expansion of [18] to a setting with possibly negative coefficients. Our work lies in the field of categorical differentiation, whose aim is to study the categorical properties of differentiation [2, 3, 8] and of Taylor expansion [25, 34, 47].

1 Partial Commutative Monoids (PCMs)

We need a suitable axiomatization of partial sums that captures sums that are typical of analysis and topology. We use the notion of *partial commutative monoids* [22]. If A is a countable set and M is a set, an A-**indexed family** of elements of M is a function

$\vec{x} : A \to M$, written $\vec{x} = (x_a)_{a\in A}$. We consider a set M with a partial function Σ called the *sum* from indexed families on M to M. An indexed family $(x_a)_{a\in A}$ is *summable* if it is in the domain of Σ, and we write its image as $\sum_{a\in A} x_a$.

▶ **Definition 1.** *For any expression e and e' involving partial sums, we write $e \rightrightarrows e'$ (or $e' \rightrightarrows e$) if whenever e is defined, then e' is also defined and $e = e'$ We write $e \leftrightrightarrows e'$ if $e \rightrightarrows e'$ and $e \rightrightarrows e'$.*

▶ **Definition 2** ([22])**.** *The tuple (M, Σ) is a* partial commutative monoid (PCM) *if M is non-empty and if the sum Σ satisfies the following axioms.*

- *(Unary) Unary sum axiom: every singleton family (x) is summable with sum x.*
- *(WPA) Weak partition associativity axiom: Let $(x_a)_{a\in A}$ be an indexed family, I countable, and $\{A_i\}_{i\in I}$ a partition of A (the A_i's can be empty). Then $\sum_{a\in A} x_a \rightrightarrows \sum_{i\in I} \left(\sum_{a\in A_i} x_a\right)$*

A PCM is strong *if it satisfies the partition associativity axiom (PA), where the implication $\rightrightarrows$ of (WPA) is replaced by an equivalence $\leftrightrightarrows$. Strong PCMs are often called Σ-monoids [21].*

Observe that the neutral element 0 is not given as part of the data, but can be defined from the axioms. Indeed, the sum over the empty family is always defined by Lemma 3.

▶ **Lemma 3.** *If $A \subseteq B$ and $(x_b)_{b\in B}$ is summable, then $(x_a)_{a\in A}$ is summable.*

▶ **Definition 4.** *Every PCM has a canonical preorder (it is not antisymmetric in general) given by $x \leq y$ if there exists z such that $x + z = y$.*

Algebraic programming semantics is often based on strong PCMs and *additive domains* [36]. An additive domain is (roughly) a *strong* PCM whose preorder of Definition 4 is antisymmetric, and where $\sum_{a\in A} x_a = \sup\left\{\sum_{a\in F} x_a \mid F \subseteq A \text{ finite}\right\}$. However, strong PCMs (M, Σ) are *positive* [22]: for all $x, y \in M$, $x + y = 0 \Rightarrow x = y = 0$, as illustrated in Examples 5 and 6. Positivity is not an issue when one is concerned with programming languages semantics. However, it does not capture partial summability in models based on traditional analysis.

▶ **Example 5.** A family of *non-negative* real numbers $(x_a \in \mathbb{R}_{\geq 0})_{a\in A}$ is summable when $D = \{\sum_{a\in F} x_a | F \text{ is a finite subset of } A\}$ has an upper bound, written $\sum_{a\in A} x_a = \sup D$. This sum defines an additive domain. A countable family of real numbers $(x_a \in \mathbb{R})_{a\in A}$ is *absolutely convergent* if the family of their absolute values $(|x_a|)_{a\in A}$ is summable. Then we can define the sum of the $(x_a)_{a\in A}$ as $\sum_{a\in A} x_a = \sum_{a\in A} |x|_a^+ - \sum_{a\in A} |x|_a^-$, where $|x|_a^+ = \max(0, x_a)$ and $|x|_a^- = \max(0, -x_a)$. The notion of absolute convergence yields a PCM, the axiom (WPA) corresponds to the Fubini-Tonelli theorem. However, this PCM is not strong: $(-1, 1)$ is summable with sum 0, $(0, 0, \ldots)$ is summable, but $(-1, 1, -1, 1, \ldots)$ is not.

▶ **Example 6.** For every monoid M, we can define the *finitary PCM* generated by M in which a family $\vec{x} = (x_a)_{a\in A}$ is summable if $\mathsf{supp}(\vec{x}) = \{a \in A \mid x_a \neq 0\}$ is finite, with sum defined as $\sum_{a\in A} x_a = \sum_{a\in \mathsf{supp}(\vec{x})} x_a$. This PCM is strong if and only if M is positive.

Properties of PCMs. First, the sum over the empty family, 0, is a neutral element.

▶ **Definition 7.** *We define the* support *of a family $\vec{x} = (x_a)_{a\in A}$ as $\mathsf{supp}(\vec{x}) = \{a \in A | x_a \neq 0\}$.*

▶ **Proposition 8** ([22, Prop. 2.4])**.** *Let $\vec{x} = (x_a)_{a\in A}$ be an indexed family, and A' be a set such that $\mathsf{supp}(\vec{x}) \subseteq A' \subseteq A$. Then $\sum_{a\in A'} x_a \leftrightrightarrows \sum_{a\in A} x_a$.*

Proof. Assume that $(x_a)_{a\in A'}$ is summable. Then for all $a \in A$, define $A_a = \{a\}$ if $a \in A'$, and $A_a = \emptyset$ otherwise. Then, $\{A_a\}_{a\in A}$ is a partition of A'. Thus, by (WPA), $\sum_{a\in A'} x_a \rightrightarrows \sum_{a\in A} \sum_{a'\in A_a} x_{a'}$. But for all $a \in A$, $\sum_{a'\in A_a} x_{a'} = x_a$, by (Unary) if $x_a \neq 0$, and by

definition of 0 if $x_a = 0$. Thus, $\sum_{a\in A'} x_a \rightrightarrows \sum_{a\in A} x_a$. Conversely, if $(x_a)_{a\in A}$ is summable then $(x_a)_{a\in A'}$ is summable by Lemma 3 because it is a subfamily of $(x_a)_{a\in A}$, and it is proved above that the sums are equal. ◀

Sums in PCMs are agnostic to the index set. Given two index sets A and B, an injection $\phi : A \hookrightarrow B$ and a family $\vec{x} = (x_a)_{a\in A}$ we define a B-indexed family $\phi^*\vec{x} = (y_b)_{b\in B}$ by $y_{\phi(a)} = x_a$, and $y_b = 0$ if $b \notin \phi(A)$. We now generalize [22, Prop. 2.3] to injections.

▶ **Proposition 9.** *For any injection $\phi : A \hookrightarrow B$ and any A-indexed family $\vec{x}$, $\phi^*\vec{x}$ is summable if and only if $\vec{x}$ is summable, and the two sums are equal.*

Proof. Let $\vec{y} = \phi^*\vec{x} = (y_b)_{b\in B}$. By definition, $\mathsf{supp}(\vec{y}) \subseteq \mathrm{Im}(\phi) \subseteq B$, so by Proposition 8, $\vec{y}$ is summable if and only if $(y_b)_{b\in \mathrm{Im}(\phi)}$ is summable, and the two sums are equal. Thus, it suffices to prove our result for all bijections to conclude that it holds for all injections. We now assume that ϕ is a bijection. Assume that $\vec{x}$ is summable. For all $b \in B$, let $A_b = \{\phi^{-1}(b)\}$. Then the $\{A_b\}_{b\in B}$ are a partition of A, so by (WPA) we have $\sum_{a\in A} x_a \rightrightarrows \sum_{b\in B}\sum_{a\in A_b} x_a$. By (Unary), $\sum_{a\in A_b} x_a = \phi^*\vec{x}(b)$. Thus, $\phi^*\vec{x}$ is summable with the same sum as $\vec{x}$. Conversely, if $\phi^*\vec{x}$ is summable then $(\phi^{-1})^*\phi^*\vec{x} = \vec{x}$ is summable. ◀

Partial commutative rigs. We introduce a PCM analogue to the notion of rigs (also called commutative semirings) [20]. They will serve as the coefficients of our web models (Part I).

▶ **Definition 10.** *A Partial Commutative Rig (PCR) is the data of $(\mathcal{R}, \Sigma, 1, \cdot)$ such that $(\mathcal{R}, \Sigma)$ is a PCM, $(\mathcal{R}, 1, \cdot)$ is a commutative monoid, and such that for all indexed families $(x_a)_{a\in A}$ and $(y_b)_{b\in B}$ of $\mathcal{R}$, $\left(\sum_{a\in A} x_a\right)\cdot\left(\sum_{b\in B} y_b\right) \rightrightarrows \sum_{a\in A, b\in B}(x_a \cdot y_b)$. A PCR is strong if the underlying PCM is strong. Strong PCR are called Σ-semirings in [45].*

▶ **Example 11.**

- Both $\mathbb{R}$ and $\mathbb{R}_{\geq 0}$ are PCR, with the PCM structure of Example 5 and with the usual multiplication.
- Every rig $(\mathcal{R}, 0, +, 1, \cdot)$ induces a PCR called the *finitary PCR* generated by $\mathcal{R}$, in which the PCM structure is the finitary PCM induced by $(\mathcal{R}, +)$ (see Example 6).
- A complete rig [32] is the same as a PCR in which sums are always defined. A *continuous rig* [33] is a complete rig in which the PCM structure is an additive domain. A standard example of complete rig is the completion of positive real numbers $\overline{\mathbb{R}}_{\geq 0} = \mathbb{R}_{\geq 0} \cup \{\omega\}$.

PCM categories. We now give a definition of categories "enriched" over PCM. This enrichment is at the core of our theory of Taylor expansion, in the same way that differential categories [2] are enriched over commutative monoids. Let $\mathcal{L}$ be a locally small category such that for all objects X, Y, $\mathcal{L}(X, Y)$ is a PCM.

▶ **Definition 12** ([22])**.** *The category $\mathcal{L}$ is a PCM category if for all $(f_a \in \mathcal{L}(X, Y))_{a\in A}$ and $(g_b \in \mathcal{L}(Y, Z))_{b\in B}$, $\left(\sum_{b\in B} g_b\right)\circ\left(\sum_{a\in A} f_a\right) \rightrightarrows \sum_{(a,b)\in A\times B}(g_b \circ f_a)$.*

This property is called the strong distributivity of the composition. By (Unary), strong distributivity implies both left and right distributivity: $\left(\sum_{b\in B} g_b\right)\circ f \rightrightarrows \sum_{b\in B}(g_b \circ f)$ and $g \circ \left(\sum_{a\in A} f_a\right) \rightrightarrows \sum_{a\in A}(g \circ f_a)$. However, left and right distributivity do not imply strong distributivity, unless the PCM is strong [22]. We need to be very careful about this subtlety.

▶ **Definition 13.** *A category $\mathcal{L}$ has zero morphisms if for all X, Y, there exists a morphism $0^{X,Y} \in \mathcal{L}(X, Y)$ such that for all $f \in \mathcal{L}(X', X)$ and $g \in \mathcal{L}(Y, Y')$, $0^{X,Y} \circ f = 0^{X',Y}$ and $g \circ 0^{X,Y} = 0^{X,Y'}$. When the category has zero morphism, we define for all object X, the Kronecker symbol $\delta^j_i \in \mathcal{L}(X, X)$ by $\delta^i_i = \mathsf{id}_X$ and $\delta^j_i = 0$ if $i \neq j$.*

It follows from left and right distributivity that a PCM category always have 0-morphisms, defined as the sum over the empty family.

Part I

A generic construction of web models

We provide in this part a unified construction of web models of linear logic, based on a generic PCR $\mathcal{R}$. This definition encompasses many web based models from the literature depending on the choice of $\mathcal{R}$ (see Example 17), and clearly identify what common patterns these models share. We give two constructions. The first one follows the standard construction of most web based models, but only works when the PCR is *strong*. The second one works on more general PCR, whose summability consists of some sort of *absolute convergence*.

Let us introduce new notations for this part. We define the Kronecker symbol: $\delta_a^b = 0$ if $a \neq b$, and $\delta_a^a = 1$. We define for all $a \in |X|$ a vector $e_a \in \mathcal{R}^{|X|}$ by $(e_a)_{a'} = \delta_a^{a'}$. For all set $|X|$ (called the *web*) and all PCR $\mathcal{R}$, we define the following *partial* maps on vectors:

- scalar product $\langle .|. \rangle_{\mathcal{R}} : \mathcal{R}^{|X|} \times \mathcal{R}^{|X|} \to \mathcal{R}$ by $\langle x|y \rangle_{\mathcal{R}} = \sum_{a \in |X|} x_a y_a$,
- matrix multiplication $_\cdot_ : \mathcal{R}^{|Y|\times|Z|} \times \mathcal{R}^{|X|\times|Y|} \to \mathcal{R}^{|X|\times|Z|}$ by $(t \cdot s)_{a,c} = \sum_{b \in |Y|} s_{a,b} \cdot t_{b,c}$,
- and the application of a matrix to a vector $_ \cdot _ : \mathcal{R}^{|X|\times|Y|} \times \mathcal{R}^{|X|} \to \mathcal{R}^{|Y|}$ (this is an instance of matrix multiplication) by $(s \cdot x)_b = \sum_{a \in |X|} s_{a,b} x_a$.

Notations on LL. Let $\mathcal{L}$ be a model of **linear logic**, more precisely a Seely category [38]. Let $(!, \mathsf{der}, \mathsf{dig})$ be the resource comonad, $\mathcal{L}_!$ the Kleisli category of $!_$. We write the composition of $f \in \mathcal{L}(X, Y)$ with $g \in \mathcal{L}(Y, Z)$ as $g \circ f$. We write the symmetric monoidal structure as $(\otimes, \lambda, \rho, \alpha, \gamma)$ where $\lambda_X : 1 \otimes X \to X$ and $\rho_X : X \otimes 1 \to X$ are the unitors, $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ is the associator, and $\gamma_{X,Y} : X \otimes Y \to Y \otimes X$ is the symmetry. We write $X \multimap Y$ the internal hom, the currying of $f : X \otimes Y \to Z$ is $\mathsf{cur}(f) : X \to (Y \multimap Z)$, and the evaluation is $\mathsf{ev} : (X \multimap Y) \otimes X \to Y$. There is a bifunctor $_ \multimap _ : \mathcal{L}^{op} \times \mathcal{L} \to \mathcal{L}$ that maps $f : X' \to X$ and $g : Y \to Y'$ to $f \multimap g = \mathsf{cur}(g \circ \mathsf{ev} \circ ((X \multimap Y) \otimes f)) : (X \multimap Y) \to (X' \multimap Y')$. We write $X : I \to \mathcal{L}$ the constant functor, where I is the category with one object and one morphism. This functor induces two functors $X \otimes _ : \mathcal{L} \to \mathcal{L}$ and $X \multimap _ : \mathcal{L} \to \mathcal{L}$. We write I-indexed product as $\&_{i \in I} X_i$, the projections as $\mathsf{p}_i : \&_{i \in I} X_i \to X_i$. We write I-indexed coproducts as $\bigoplus_{i \in I} X_i$, the injections as $\mathsf{in}_i : X_i \to \bigoplus_{i \in I} X_i$. We write the paring of a family $f_i : X \to Y_i$ as $\langle f_i \rangle_{i \in I} : X \to \&_{i \in I} Y_i$, and the co-pairing of $f_i : X_i \to Y$ as $[f_i]_{i \in I} : \bigoplus_{i \in I} X_i \to Y$. A finite **multiset** over a set A is a function $m : A \to \mathbb{N}$ with finite support. Let $\mathcal{M}_{\mathrm{fin}}(A)$ be the set of all finite multisets over A. We write $m_1 + m_2$ the pointwise sum of multisets. We also use the notation $[a_1, \ldots, a_n]$ for the multiset that maps a to the size of $\{i \mid a = a_i\}$.

2 Web models based on strong PCMs

2.1 Orthogonality and objects

First we assume that $\mathcal{R}_+$ is a strong PCR, and we consider $\mathfrak{B} \subseteq \mathcal{R}_+$. We assume that $\mathfrak{B}$ is *downward closed*: if $x \leq y$ and $y \in \mathfrak{B}$ then $x \in \mathfrak{B}$. This assumption is not necessary to build the model, but it will be necessary to ensure that the construction yields PCM categories. We define an *orthogonality relation* $x \perp y$ on vectors and the *orthogonal* of a set $F \subseteq \mathcal{R}_+^{|X|}$ as:

$$x \perp y \text{ if } \langle x|y \rangle_{\mathcal{R}} \text{ is defined and belongs to } \mathfrak{B} \qquad F^\perp = \{x' \in \mathcal{R}_+^{|X|} \mid x \perp x'\}.$$

This construction has the usual properties of an orthogonality construction: $F \subseteq F^{\perp\perp}$, $(F \subseteq G) \Rightarrow (G^\perp \subseteq F^\perp)$, and $F^{\perp\perp\perp} = F^\perp$. Different choices of $\mathcal{R}_+$ and $\mathfrak{B}$ yield the usual definitions of orthogonality in various web models (see Example 17).

▶ **Definition 14.** *A set of vectors $F \subseteq \mathcal{R}_+^{|X|}$ is a covering if for all $a \in |X|$, there exists $x \in F$ such that its a component x_a is invertible.*

▶ **Definition 15.** *An $\mathcal{R}_+$-space is a pair $X = (|X|, \mathcal{A}X)$ where $|X|$ is a set and $\mathcal{A}X \subseteq \mathcal{R}_+^{|X|}$ is a set of vectors such that $\mathcal{A}X^{\perp\perp} = \mathcal{A}X$, and such that both $\mathcal{A}X$ and $\mathcal{A}X^{\perp}$ are coverings. The dual of an $\mathcal{R}_+$-space X is defined as the $\mathcal{R}_+$-space $X^{\perp} = (|X|, \mathcal{A}X^{\perp})$.*

The downward closure of $\mathfrak{B}$ induces the downward closure of $\mathcal{R}_+$-spaces, for the pointwise preorder on $\mathcal{R}_+^{|X|}$ induced by the preorder on $\mathcal{R}_+$ given in Definition 4.

▶ **Lemma 16** (Downward closure)**.** *For all $x, y \in \mathcal{R}_+^{|X|}$ such that $x \leq y$, $y \in \mathcal{A}X \Rightarrow x \in \mathcal{A}X$.*

▶ **Example 17.** This definition and the PCRs of Example 11 describe many web models.

- Relational model [13, App. A]: take $\mathcal{R}_+$ to be the complete boolean rig $\{0, 1\}$ (the infinite sum $1 + 1 + \ldots$ is defined and equal to 1), and $\mathfrak{B} = \mathcal{R}_+$.
- Weighted relational model [33]: take $\mathcal{R}_+$ to be any continuous rig (or simply complete, see [32]), and $\mathfrak{B} = \mathcal{R}_+$.
- Probabilistic coherence spaces [11]: take $\mathcal{R}_+ = \mathbb{R}_{\geq 0}$ (recall Example 5) and $\mathfrak{B} = [0, 1]$.
- Coherence spaces [19]: take $\mathcal{R}_+ = \{0, \omega\}$ with $\omega + \omega$ undefined, and $\mathfrak{B} = \{0, \omega\}$.
- Finiteness spaces (relational version) [13]: take $\mathcal{R}_+$ to be the finitary PCM induced by the finite boolean rig $\{0, 1\}$ (the infinite sum $1 + 1 + \ldots$ is not defined), and $\mathfrak{B} = \mathcal{R}_+$.
- Köthe spaces (positive version) [12]: take $\mathcal{R}_+ = \mathbb{R}_{\geq 0}$ and $\mathfrak{B} = \mathbb{R}_{\geq 0}$.

Different choices of PCR describe different computational properties: the countable non-determinism of the (weighted) relational model, the randomness of probabilistic coherence spaces, the determinism for coherence spaces, and the finite non-determinism for finiteness spaces. There are however some web models that cannot be described by an orthogonality relation, for instance the category of non-uniform coherence spaces [5].

2.2 Tensor product of objects and morphisms

We can now build the category of $\mathcal{R}_+$-spaces following the standard procedure of web based models: we first build the tensor product of objects, and we deduce the morphisms by duality. For all vectors $x \in \mathcal{R}_+^{|X|}$ and $y \in \mathcal{R}_+^{|Y|}$, define $x \otimes y \in \mathcal{R}_+^{|X| \times |Y|}$ by $(x \otimes y)_{a,b} = x_a y_b$. For all sets $F \subseteq \mathcal{R}_+^{|X|}$ and $G \subseteq \mathcal{R}_+^{|Y|}$, we set $F \otimes G = \{x \otimes y \mid x \in F, y \in G\}$. The *tensor* $X \otimes Y$ of two $\mathcal{R}_+$-spaces X and Y is defined as

$$|X \otimes Y| = |X| \times |Y| \qquad \mathcal{A}(X \otimes Y) = (\mathcal{A}X \otimes \mathcal{A}Y)^{\perp\perp}.$$

The closure under double orthogonality is crucial to ensure that this is an $\mathcal{R}_+$-space. We now define the $\mathcal{R}_+$-space of linear maps between $\mathcal{R}_+$-spaces by duality: $X \multimap Y = (X \otimes Y^{\perp})^{\perp}$.

▶ Remark 18. $\mathcal{A}(X \multimap Y) = \mathcal{A}(X \otimes Y^{\perp})^{\perp} = (\mathcal{A}X \otimes \mathcal{A}Y^{\perp})^{\perp\perp\perp} = (\mathcal{A}X \otimes \mathcal{A}Y^{\perp})^{\perp}$. Thus, a matrix $s \in \mathcal{R}^{|X \multimap Y|}$ belongs to $\mathcal{A}(X \multimap Y)$ if and only if $s \perp (x \otimes y')$ for all $x \in \mathcal{A}X, y' \in \mathcal{A}Y^{\perp}$. This predual characterization of $X \multimap Y$ is the backbone of web semantics.

▶ **Theorem 19.** *For all $s \in \mathcal{A}(X \multimap Y)$ and $t \in \mathcal{A}(Y \multimap Z)$, $t \cdot s$ is well-defined and belongs to $\mathcal{A}(X \multimap Z)$. Furthermore, the identity matrix $\mathsf{id}_X \in \mathcal{R}_+^{|X \multimap X|}$ belongs to $\mathcal{A}(X \multimap X)$. Thus, we can define the category $\mathbf{Web}_{\mathcal{R}_+}$ whose objects are the $\mathcal{R}_+$-spaces, whose morphisms are $\mathbf{Web}_{\mathcal{R}_+}(X, Y) = \mathcal{A}(X \multimap Y)$, and whose composition is given by matrix multiplication.*

Proof sketch of Theorem 19. The proof (see Appendix A) uses the predual characterization described in Remark 18. Now, observe that $\langle s \mid x \otimes y' \rangle$ can be reordered as follows.

▶ **Lemma 20.** *For all $s \in \mathcal{R}_+^{|X|\times|Y|}$, $x \in \mathcal{R}_+^{|X|}$ and $y \in \mathcal{R}_+^{|Y|}$,*

- *$\langle s \cdot x \mid y\rangle \rightrightarrows \langle s \mid x \otimes y\rangle$. Furthermore, if $s \cdot x$ is defined, then $\langle s \cdot x \mid y\rangle \rightleftarrows \langle s \mid x \otimes y\rangle$.*
- *$\langle s^\perp \cdot y \mid x\rangle \rightrightarrows \langle s \mid x \otimes y\rangle$. Furthermore, if $s^\perp \cdot y$ is defined, then $\langle s^\perp \cdot y \mid x\rangle \rightleftarrows \langle s \mid x \otimes y\rangle$.*

Where $s^\perp \in \mathcal{R}_+^{|Y|\times|X|}$ is the transpose of s, $(s^\perp)_{b,a} = s_{a,b}$

Unfortunately, the equivalences in Lemma 20 requires $s\cdot x$ and $s^\perp\cdot y$ to be defined. Fortunately, we can ensure that this condition often holds, by Lemma 21 below.

▶ **Lemma 21** (Covering principle)**.** *Let Q be a covering of $\mathcal{R}_+^{|Y|}$. Let $x \in \mathcal{R}_+^{|X|}$ and $s \in \mathcal{R}_+^{|X\multimap Y|}$ such that $\langle s \mid x \otimes y\rangle$ is well-defined for all $y \in Q$. Then $s \cdot x$ is well-defined.*

Similarly, let P be a covering of $\mathcal{R}_+^{|X|}$. Let $y \in \mathcal{R}_+^{|Y|}$ and $s \in \mathcal{R}_+^{|X\multimap Y|}$ such that $\langle s \mid x \otimes y\rangle$ is well-defined for all $x \in P$. Then $s^\perp \cdot y$ is well-defined.

Used together, Lemmas 20 and 21 give us a convenient characterization of $\mathcal{A}(X \multimap Y)$. This characterization gives a unified account of seemingly different characterizations that appears in all web models, such as probabilistic coherence spaces [11, Lemma 3], finiteness spaces [13, Lemma 3], and even the historical model of coherence spaces [19, Def. 3.6].

▶ **Proposition 22.** *Let $s \in \mathcal{R}_+^{|X\multimap Y|}$. The following are equivalent:*

1. *$s \in \mathcal{A}(X \multimap Y)$;*
2. *$s^\perp \in \mathcal{A}(Y^\perp \multimap X^\perp)$;*
3. *$\forall x \in \mathcal{A}X$, $s \cdot x$ is defined and belongs to $\mathcal{A}Y$;*
4. *$\forall y' \in \mathcal{A}Y^\perp$, $s^\perp \cdot y'$ is defined and belongs to $\mathcal{A}X^\perp$.*

The proof of Theorem 19 uses this characterization. First, we prove that $t \cdot s$ is well-defined with an argument similar to that of Lemma 21. Then, we prove that $(t \cdot s) \cdot x \rightleftarrows t \cdot (s \cdot x)$, and we conclude that $t \cdot s \in \mathcal{A}(X \multimap Z)$ by applying Proposition 22 twice. ◀

To wrap up this section, we further improve the predual characterization of $X \multimap Y$ given in Remark 18 and Proposition 22. This new predual characterization captures in a unified way various proof techniques that appear throughout the literature on web models.

▶ **Theorem 23** (Predual characterization of morphisms)**.** *Let X and Y be two $\mathcal{R}_+$-spaces. Let $P \subseteq \mathcal{R}_+^{|X|}$ and $Q \subseteq \mathcal{R}_+^{|Y|}$ be two coverings such that $P^{\perp\perp} = \mathcal{A}X$ and $Q^\perp = \mathcal{A}Y$. Then for all $s \in \mathcal{R}_+^{|X\multimap Y|}$, the following are equivalent.*

1. *$s \in \mathcal{A}(X \multimap Y)$;*
2. *$s \in (P \otimes Q)^\perp$;*
3. *for all $x \in P$, $s \cdot x$ is defined and belongs to $\mathcal{A}Y$;*
4. *for all $y' \in Q$, $s^\perp \cdot y'$ is defined and belongs to $\mathcal{A}X^\perp$.*

The proof of Theorem 23 relies heavily on Lemmas 20 and 21 (see Appendix A). The following result immediately follows from Theorem 23, using that $X \otimes Y = (X \multimap Y^\perp)^\perp$.

▶ **Corollary 24.** *Let $P \subseteq \mathcal{R}_+^{|X|}$ and $Q \subseteq \mathcal{R}_+^{|Y|}$ be two coverings such that $P^{\perp\perp} = \mathcal{A}X$ and $Q^{\perp\perp} = \mathcal{A}Y$. Then $\mathcal{A}(X \otimes Y) = (P \otimes Q)^{\perp\perp}$.*

2.3 The linear logical structure of web models

We describe here the objects of the linear logical structure of $\mathbf{Web}_{\mathcal{R}_+}$. The rest of the structure (symmetric monoidal structure, closure, resource comonad, and Seely isomorphisms) is the usual structure of web based models (summarized in Figure 1).

The predual characterization of Theorem 23 ensures that all the matrices describing the LL structure (in Figure 1) are morphisms. We define the exponential of an $\mathcal{R}_+$-space X by:

$$|!X| = \{m \in \mathcal{M}_{\text{fin}}(|X|) \mid \exists x \in \mathcal{A}X \text{ such that } \forall a \in m, x_a \text{ is invertible}\}$$

$$\mathcal{A}(!X) = \left\{x^! \mid x \in \mathcal{A}X\right\}^{\perp\perp} \text{ where } (x^!)_{[a_1,\dots,a_n]} = x_{a_1} \cdots x_{a_n}$$

Monoidal structure: $1 = (\{*\}, \{e_*\}^{\perp\perp})$ and for all $s \in \mathcal{A}(X_1 \multimap Y_1)$ and $s' \in \mathcal{A}(X_2 \multimap Y_2)$, the tensor $s \otimes s' \in \mathcal{R}_+^{|X_1 \otimes X_2 \multimap Y_1 \otimes Y_2|}$ is defined by $(s \otimes s')_{(a,a'),(b,b')} = s_{a,b} s'_{a',b'}$ so that $(s \otimes s') \cdot (x \otimes x') = (s \cdot x) \otimes (s' \cdot x')$

$$\begin{array}{rll}
\lambda \in \mathcal{R}_+^{|1 \otimes X \multimap X|} & \lambda_{(*,a),a'} = \delta_a^{a'} & \lambda \cdot (r \otimes x) = rx \\
\rho \in \mathcal{R}_+^{|X \otimes 1 \multimap X|} & \rho_{(a,*),a'} = \delta_a^{a'} & \rho \cdot (x \otimes r) = rx \\
\alpha \in \mathcal{R}_+^{|(X \otimes Y) \otimes Z \multimap X \otimes (Y \otimes Z)|} & \alpha_{((a,b),c),(a',(b',c'))} = \delta_a^{a'} \delta_b^{b'} \delta_c^{c'} & \alpha \cdot ((x \otimes y) \otimes z) = x \otimes (y \otimes z) \\
\gamma \in \mathcal{R}_+^{|X \otimes Y \multimap Y \otimes X|} & \gamma_{(a,b),(b',a')} = \delta_a^{a'} \delta_b^{b'} & \gamma \cdot (x \otimes y) = y \otimes x
\end{array}$$

Closed structure: for all $s \in \mathcal{A}(X \otimes Y \multimap Z)$

$$\begin{array}{rll}
\mathsf{ev} \in \mathcal{R}_+^{|((X \multimap Y) \otimes X) \multimap Y|} & \mathsf{ev}_{((a,b),a'),b'} = \delta_a^{a'} \delta_b^{b'} & \mathsf{ev} \cdot (s \otimes x) = s \cdot x \\
\mathsf{cur}(s) \in \mathcal{R}_+^{|X \multimap (Y \multimap Z)|} & \mathsf{cur}(s)_{a,(b,c)} = s_{(a,b),c} & (\mathsf{cur}(s) \cdot x) \cdot y = s \cdot (x \otimes y)
\end{array}$$

Exponential: for all $s \in \mathcal{A}(X \multimap Y)$

$$\begin{array}{rll}
!s \in \mathcal{R}_+^{|!X \multimap !Y|} & (!s)_{m,[b_1,\ldots,b_n]} = \sum_{\substack{a_1,\ldots,a_n \in |X| \\ \text{s.t. } m = [a_1,\ldots,a_n]}} s_{a_1,b_1} \cdots s_{a_n,b_n} & !s \cdot x^! = (s \cdot x)^! \\
\mathsf{der} \in \mathcal{R}_+^{|!X \multimap X|} & \mathsf{der}_{m,a} = \delta_a^{[a]} & \mathsf{der} \cdot x^! = x \\
\mathsf{dig} \in \mathcal{R}_+^{|!!X \multimap !X|} & \mathsf{dig}_{[m_1,\ldots,m_k],m} = \delta_m^{m_1+\cdots+m_n} & \mathsf{dig} \cdot (x^!)^! = x^!
\end{array}$$

Seely isomorphisms: Define $k \cdot [a_1, \ldots, a_n] = [(k, a_1), \ldots, (k, a_n)]$. Observe that every $m \in |!(X \,\&\, Y)|$ can be written uniquely as $m = 1 \cdot m_1 + 2 \cdot m_2$ with $m_1 \in |!X|$ and $m_2 \in |!Y|$ (even when taking into account that $|!X|$ is not the set of all multisets in general). Furthermore, recall that $\top = (\emptyset, \{0\})$ so $!\top = \left(\{[]\}, \{0^!\}^{\perp\perp}\right) = \left(\{[]\}, \{e_{[]}\}^{\perp\perp}\right)$.

$$\begin{array}{rll}
\mathsf{m}^0 \in \mathcal{R}_+^{|1 \multimap !\top|} & {\mathsf{m}^0}_{*,[]} = 1 & \mathsf{m}^0 \cdot r = r \\
(\mathsf{m}^0)^{-1} \in \mathcal{R}_+^{|!\top \multimap 1|} & (\mathsf{m}^0)^{-1}_{[],*} = 1 & (\mathsf{m}^0)^{-1} \cdot r = r \\
\mathsf{m}^2 \in \mathcal{R}_+^{|(!X \otimes !Y) \multimap !(X \& Y)|} & {\mathsf{m}^2}_{(m_1,m_2),m} = \delta_m^{1 \cdot m_1 + 2 \cdot m_2} & \mathsf{m}^2 \cdot (x^! \otimes y^!) = \langle x, y \rangle^! \\
(\mathsf{m}^2)^{-1} \in \mathcal{R}_+^{|!(X \& Y) \multimap (!X \otimes !Y)|} & (\mathsf{m}^2)^{-1}_{m,(m_1,m_2)} = \delta_m^{1 \cdot m_1 + 2 \cdot m_2} & (\mathsf{m}^2)^{-1} \cdot \langle x, y \rangle^! = x^! \otimes y^!
\end{array}$$

Figure 1 LL structure of $\mathbf{Web}_{\mathcal{R}_+}$.

Observe that $|!X| \neq \mathcal{M}_{\text{fin}}(|X|)$ in general because we want $\mathcal{A}(!X)$ to be a covering. Indeed, in coherence spaces the web of $!X$ is the set of multisets whose support is a clique.

By design, the category is $*$-autonomous, with dualizing object $\perp = 1^\perp$. It has products and coproducts. The webs are given by $\left|\&_{i \in I} X_i\right| = \left|\bigoplus_{i \in I} X_i\right| = \bigcup_{i \in I} \{i\} \times X_i$. We can define for all $i \in I$ injections $\mathsf{in}_i \in \mathcal{R}_+^{\left|X_i \multimap \left(\bigoplus_{i \in I} X_i\right)\right|}$ and projections $\mathsf{p}_i \in \mathcal{R}_+^{|(\&_{i \in I} X_i) \multimap X_i|}$

by $(\mathsf{in}_i)_{a,(j,a')} = (\mathsf{p}_i)_{(j,a),a'} = \delta_i^j \delta_a^{a'}$. We can then define

$$\mathcal{A}\Big(\mathop{\&}_{i\in I} X_i\Big) = \Big\{\mathsf{in}_i \cdot x_i \mid x_i \in \mathcal{A}X_i{}^{\perp}\Big\}^{\perp} \qquad \mathcal{A}\Big(\bigoplus_{i\in I} X_i\Big) = \{\mathsf{in}_i \cdot x_i \mid x_i \in \mathcal{A}X_i\}^{\perp\perp},$$

and we can check that these are products and coproducts. Observe in particular that for all $x \in \mathcal{R}_+^{\left|\&_{i\in I} X_i\right|}$, $\langle \mathsf{p}_i \cdot x \mid x'\rangle \rightleftarrows \langle x \mid \mathsf{in}_i \cdot x'\rangle$ so $x \in \mathcal{A}\left(\&_{i\in I} X_i\right)$ if and only if for all $i \in I$, $\mathsf{p}_i \cdot x \in \mathcal{A}X_i$. In particular, $\mathcal{A}\left(\bigoplus_{i\in I} X_i\right) \subseteq \mathcal{A}\left(\&_{i\in I} X_i\right)$, but this is not an equality in general. The initial and terminal object is $\top = (\emptyset, \{0\})$.

▶ Remark 25. We conjecture that the exponential we describe here is always Lafont [31]. All the web models given in Example 17 are known to be Lafont, except Köthe spaces for which the question is still open. However, the proof changes drastically depending on the model: the proof that probabilistic coherence spaces are Lafont [9] uses a generic formula that fails in finiteness spaces [39].

3 Web models based on absolute convergence

We now describe web models based on PCRs that are not strong, but whose summability behaves similarly to the absolute convergence on real numbers (Example 5). We give two examples: Köthe spaces [12], and the category of modules over a finiteness space [13].

▶ **Definition 26.** *A PCR $\mathcal{R}$ is absolute if there exists a strong PCR $\mathcal{R}_+$ and a function $|.| : \mathcal{R} \to \mathcal{R}_+$ (called the absolute value) such that*

- *$|x| = 0$ if and only if $x = 0$;*
- *$|.|$ is a monoid morphism for the multiplicative part: $|xy| = |x||y|$ and $|1| = 1$;*
- *Triangle inequality: $(x_a)_{a\in A}$ is summable in $\mathcal{R}$ if and only if $(|x_a|)_{a\in A}$ is summable in $\mathcal{R}_+$, and $|\sum_{a\in A} x_a| \leq \sum_{a\in A} |x_a|$.*

▶ **Example 27.** The PCR of real numbers is absolute, taking $\mathcal{R}_+ = \mathbb{R}_{\geq 0}$ and $|.|$ to be the regular absolute value. The finitary PCR induced by a rig (Example 11) is also absolute, taking $\mathcal{R}_+$ to be the finitary PCR induced by the boolean rig, and $|x| = \delta_x^1$.

We generalize to inequalities the notations of Definition 1. For example, the triangle inequality stated in Definition 26 above is described by the equation $|\sum_{a\in A} x_a| \lesssim \sum_{a\in A} |x_a|$. The absolute value $|.|$ extends to a function $\mathsf{abs} : \mathcal{R}^{|X|} \to \mathcal{R}_+^{|X|}$ for all web $|X|$, defined by $\mathsf{abs}(x)_a = |x_a|$ for all $a \in |X|$. By triangle inequality, for all $s \in \mathcal{R}^{|X \multimap Y|}$ and $t \in \mathcal{R}^{|Y \multimap Z|}$, $\mathsf{abs}(t \cdot s) \lesssim \mathsf{abs}(t) \cdot \mathsf{abs}(s)$. Similarly, for all $x, y \in \mathcal{R}^{|X|}$, $|\langle x \mid y\rangle| \lesssim \langle \mathsf{abs}(x) \mid \mathsf{abs}(y)\rangle$.

▶ **Definition 28.** *For all $\mathcal{R}_+$-space X, let $\mathcal{R}\langle X\rangle = \left\{x \in \mathcal{R}^{|X|} \mid \mathsf{abs}(x) \in \mathcal{A}X\right\}$. We define the category $\mathbf{Web}_{\mathcal{R}}$ whose objects are the $\mathcal{R}_+$-spaces, whose morphisms from X to Y are the matrices in $\mathcal{R}\langle X \multimap Y\rangle$, and whose composition is matrix multiplication.*

For all $s \in \mathcal{R}\langle X \multimap Y\rangle$ and $t \in \mathcal{R}\langle Y \multimap Z\rangle$, we have that $\mathsf{abs}(t \cdot s) \lesssim \mathsf{abs}(t) \cdot \mathsf{abs}(s) \in \mathcal{A}(X \multimap Z)$ so $t \cdot s$ is well-defined and belongs to $\mathcal{R}\langle X \multimap Z\rangle$ by downward closure of $\mathcal{A}(X \multimap Z)$ (Lemma 16). We can check that $\mathbf{Web}_{\mathcal{R}}$ is a model of linear logic, with the same structure as the one described in Section 2.3. These matrices are all morphisms in $\mathbf{Web}_{\mathcal{R}}$, because $\mathsf{abs}(x \otimes y) = \mathsf{abs}(x) \otimes \mathsf{abs}(y)$, $\mathsf{abs}\left(x^!\right) = \mathsf{abs}(x)^!$ and $\mathsf{abs}(!s) \lesssim !\mathsf{abs}(s)$.

▶ **Example 29.** Definition 28 captures the two seminal models of the differential λ-calculus.

- **Finiteness spaces.** The category of modules over a finiteness space [13, Sec. 4] is an instance of our Definition 28 in which $\mathcal{R}$ is a finitary PCR.

- **Köthe spaces.** Köthe spaces are defined in [12] a bit differently, by a generalization of the orthogonality relation given in Section 2 and of the notion of $\mathcal{R}_+$-space given in Definition 15 to the absolute PCR of real numbers. In the end though, by [12, Prop. 2.12] the category of Köthe spaces coincides with our definition of $\mathbf{Web}_{\mathbb{R}}$. In fact, we can check that a Köthe space $X = (|X|, \mathcal{E}X)$ in the sense of [12, Def. 2.8] is exactly the same as the module $\mathbb{R}\langle X'\rangle$ where X' is the $\mathbb{R}_{\geq 0}$-space given by $X' = (|X|, \{\mathsf{abs}(x) \mid x \in \mathcal{E}X\})$.

4 Summability in web models

We prove in this section that both the category $\mathbf{Web}_{\mathcal{R}_+}$ defined in Section 2 and the category $\mathbf{Web}_{\mathcal{R}}$ defined in Section 3 have a PCM structure on their morphisms. In fact, we prove that for all $\mathcal{R}_+$-space X, both $\mathcal{A}X$ and $\mathcal{R}\langle X\rangle$ have a PCM structure. We start with $\mathcal{A}X$.

▶ **Definition 30.** *Let X be an $\mathcal{R}_+$-space. A family $(x(i))_{i\in I}$ of elements of $\mathcal{A}X$ is summable if their pointwise sum $\left(\sum_{i\in I} x(i)\right)_a = \sum_{i\in I} x(i)_a$ is defined and belong to $\mathcal{A}X$.*

▶ **Proposition 31.** *Let X be an $\mathcal{R}_+$-space. Then $\mathcal{A}X$ is a strong PCM.*

Proof. It immediately follows from (PA) on $\mathcal{R}_+$ that for all family $(x(i))_{i\in I}$ of $\mathcal{R}_+^{|X|}$ and all partition $\{I_k\}_{k\in K}$ of I, $\sum_{i\in I} x(i) \leftrightarroweq \sum_{k\in K}\sum_{i\in I_k} x(i)$. To conclude, we still need to ensure that whenever $(x(i))_{i\in I}$ is summable in $\mathcal{A}X$, then for all $k \in K$, $(x(i))_{i\in I_k}$ is summable in $\mathcal{A}X$. This is an immediate consequence of the downward closure of $\mathcal{A}X$ (Lemma 16). ◀

▶ Remark 32. If the set $\mathfrak{B}$ used to define orthogonality is stable under multiplication, then $\mathfrak{B}$ is a strong PCR and $\mathcal{R}_+$-spaces are $\mathfrak{B}$-modules with orthogonal basis [45].

Similarly to morphisms, summability can be characterized in terms of predual. The proof is very similar to the proof of Lemma 21 and of Theorem 23.

▶ **Proposition 33.** *Let P be a covering of $\mathcal{R}_+^{|X|}$ such that $P^\perp = \mathcal{A}X$. Then a family $(x(i))_{i\in I}$ of $\mathcal{A}X$ is summable if and only if for all $x' \in P$, $\left\langle \sum_{i\in I} x(i) \;\middle|\; x' \right\rangle$ is defined and is in $\mathfrak{B}$.*

We now define the summability in $\mathcal{R}\langle X\rangle$.

▶ **Definition 34.** *Let X be an $\mathcal{R}_+$-space. A family $(x(i))_{i\in I}$ of elements of $\mathcal{R}\langle X\rangle$ is summable if $(\mathsf{abs}(x(i)))_{i\in I}$ is summable in $\mathcal{A}X$ (in the sense of Definition 30).*

By triangle inequality, $\mathsf{abs}\left(\sum_{i\in I} x(i)\right) \leftrightarrowless \sum_{i\in I} \mathsf{abs}(x(i)) \in \mathcal{A}X$ so the pointwise sum $\sum_{i\in I} x(i)$ is well-defined and belongs to $\mathcal{R}\langle X\rangle$ by downward closure of $\mathcal{A}X$.

▶ **Proposition 35.** *The pointwise sum defined above is a PCM on $\mathcal{R}\langle X\rangle$.*

Proof. Let $(x(i))_{i\in I}$ be summable in $\mathcal{R}\langle X\rangle$, and $\{I_k\}_{k\in K}$ be a partition of I. For all $k \in K$, $(\mathsf{abs}(x(i)))_{i\in I_k}$ is summable in $\mathcal{A}X$ by Lemma 3. Thus, $(x(i))_{i\in I_k}$ is summable in $\mathcal{R}\langle X\rangle$:

$$\sum_{k\in K} \mathsf{abs}\left(\sum_{i\in I_k} x(i)\right) \overset{(\circ 1)}{\leftrightarrowless} \sum_{k\in K}\sum_{i\in I_k} \mathsf{abs}(x(i)) \overset{(\circ 2)}{\leftrightarrow} \sum_{a\in A} \mathsf{abs}(x_a) \in \mathcal{A}X$$

where ($\circ$1) holds by triangle inequality and ($\circ$2) by (PA) on the strong PCM $\mathcal{A}X$. We conclude that $(\sum_{i\in I_k} x(i))_{k\in K}$ is summable in $\mathcal{R}\langle X\rangle$ by downward closure of $\mathcal{A}X$. ◀

Part II

Taylor expansion

We generalize coherent Taylor expansion [18] to models of LL enriched over not necessarily strong PCMs (Definition 12), that subsume the monoids of differential categories [2]. First, let us give a bird's-eye view of the theory developed in [18]. A *summability structure* on $\mathcal{L}$ is, in essence, a functor S with projections $\pi_i : \mathsf{S} \Rightarrow \mathsf{Id}$ such that $(f_i : X \to Y)_{i\in\mathbb{N}}$ is summable if and only if there exists $f : X \to \mathsf{S}Y$ such that $\pi_i \circ f = f_i$.

▶ Remark 36. Defining summability by an endofunctor is standard in algebraic programming semantics. However, the object $\mathsf{S}X$ is often a coproduct, as in partially additive categories [36]. Summability structures are more general, and enjoy an epi-mono factorization (see Appendix B): $\bigoplus_{i\in\mathbb{N}} X \xrightarrow{e} \mathsf{S}X \xrightarrow{m} \prod_{i\in\mathbb{N}} X$. When sums are total, this factorization collapses into an equality, and thus the category has *countable biproducts*. This collapse is an infinitary counterpart of a well known result on finite biproducts [35, Chap. 8].

The axioms of (strong) PCMs then endow the summability structure S with a canonical *bimonad* structure (a monad and a comonad structure that interact nicely) [40]. Furthermore, the compatibility between the PCM structure of $\mathcal{L}$ and its LL structure boils down to categorical structures on the monad S: a lax monoidal structure (wrt $\otimes$), a strong monoidal structure (wrt &), and an invertible pointwise structure $\mathsf{S}(X \multimap Y) \to (X \multimap \mathsf{S}Y)$ [29].

In quantitative semantics, morphisms in the Kleisli category of the resource comonad are seen as analytic maps. Taylor expansion is thus described as a functor T on $\mathcal{L}_!$ [18]. The action on object $\mathsf{T}X = \mathsf{S}X$ ensures that all sums in the Taylor expansion are well-defined. The Taylor expansion $\mathsf{T}f \in \mathcal{L}_!(\mathsf{S}X, \mathsf{S}Y)$ of $f \in \mathcal{L}_!(X, Y)$ is intuitively the function:

$$\mathsf{T}f : (x_i)_{i\in\mathbb{N}} \mapsto \left(\sum_{1\leq k\leq j} \sum_{i_1+\cdots+i_k=j} \frac{1}{k!} f^{(k)}(x_0) \cdot (x_{i_1}, \ldots, x_{i_k}) \right)_{j\in\mathbb{N}}. \tag{1}$$

The j-th coefficient of $\mathsf{T}f$ corresponds to the degree j component of the Taylor expansion of $f\left(\sum_{i\in\mathbb{N}} x_i\varepsilon^i\right)$, where ε is a formal variable reminiscent of dual numbers [44]. In particular,

$$\mathsf{T}f(x, u, 0, \ldots) = \left(\frac{1}{j!} f^{(j)}(x) \cdot (u, \ldots, u) \right)_{j\in\mathbb{N}}$$

so $\mathsf{T}f$ captures the Taylor expansion of f. The intuitive Equation (1) is validated by web model examples in [18]. Furthermore, this equation is used in [47] to build T directly in cartesian differential categories [3], even when there is no LL structure.

Let $\mathsf{Der} : \mathcal{L} \to \mathcal{L}_!$ be the canonical functor that maps $f \in \mathcal{L}(X, Y)$ to $f \circ \mathsf{der} \in \mathcal{L}(!X, Y)$. Now, observe that for all $f \in \mathcal{L}(X, Y)$, $\mathsf{Der}\, f \in \mathcal{L}(!X, Y)$ is a linear map, so we should have that $\mathsf{T}(\mathsf{Der}\, f)(x_i)_{i\in\mathbb{N}} = (\mathsf{Der}\, f(x_i))_{i\in\mathbb{N}}$, thus $\mathsf{T}(\mathsf{Der}\, f) = \mathsf{Der}\,(\mathsf{S}f)$. We say that T is an *extension* of S to $\mathcal{L}_!$ [42]. Such an extension boils down to a *distributive law* [42] between the functor S and the comonad $!_$. It consists of a natural transformation $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$ subject to a compatibility condition with der and dig. Additional compatibility conditions between ∂ and the structure on S (bimonad structure, monoidal structure with regard to $\otimes$ and &, and projection π_0) ensure that the structure on S extends to T. This structure on T describes with categorical equations the standard properties of the differential calculus [47].

Summability is often given by $\mathsf{S} = \mathbb{D} \multimap _$, with $\mathbb{D} = \&_{i\in I} 1$ (see [18]). The object $\mathbb{D}$ has a bimonoid structure that is *mate* [24] to the bimonad structure of S (see Figure 2). The mate isomorphism implies that the existence of the distributive law $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$ is equivalent to the existence of a !-coalgebra $\widetilde{\partial} \in \mathcal{L}(\mathbb{D}, !\mathbb{D})$ compatible with the bimonoid structure of $\mathbb{D}$.

Contribution. We adapt [18] to a setting where PCMs are not necessarily strong. In Section 5, we present summability structures in the absence of (PA). In Section 6, we replace the object $\mathbb{D} = \&_{i\in I} 1$ as it does not account for representable summability in Köthe spaces. In Section 7, we show that all web models defined in Part I can be captured by this theory.

5 Taylor expansion in PCM categories

Summability structure. We generalize the summability structure of [18] to PCM categories.

▶ **Definition 37.** *A* summability structure *on a category $\mathcal{L}$ is a tuple $(\mathsf{S}, (\pi_i)_{i\in\mathbb{N}}, \sigma, 0)$ where*

- *(S-fun): S is a functor on $\mathcal{L}$, and $\pi_i, \sigma : \mathsf{S} \Rightarrow \mathsf{Id}$ are natural transformations called respectively projections and sum, and 0 is a family of zero morphisms $0^{X,Y} \in \mathcal{L}(X,Y)$.*
- *(S-proj) The π_i are jointly monic: $\forall f, g \in \mathcal{L}(X, \mathsf{S}Y)$, $(\forall i \in \mathbb{N}, \pi_i \circ f = \pi_i \circ g) \implies f = g$.*
- *(S-sum) For all objects X, Y, there exists a PCM structure on $\mathcal{L}(X,Y)$ such that*
  - *(S-sum-1) A family $(f_{i_1,\dots,i_n})_{i_1,\dots,i_n\in\mathbb{N}}$ over $\mathcal{L}(X,Y)$ is summable if and only if there exists $f \in \mathcal{L}(X, \mathsf{S}^n Y)$ such that $\pi_{i_1} \circ \cdots \circ \pi_{i_n} \circ f = f_{i_1,\dots,i_n}$.*
  - *(S-sum-2) If $(f_{i_1,\dots,i_n})_{i_1,\dots,i_n\in\mathbb{N}}$ is summable then $\sum_{i_1,\dots,i_n\in\mathbb{N}} f_{i_1,\dots,i_n} = \overbrace{\sigma \circ \cdots \circ \sigma}^{n} \circ f$.*
  - *(S-sum-3) The sum over the empty family is equal to the zero morphism 0.*

The axiom (S-proj) ensures that the object $\mathsf{S}X$ describes $\mathbb{N}$-indexed families over X. It is similar to a product, but the pairing only exists if the family is summable by (S-sum).

▶ **Definition 38.** *By (S-proj), the morphism f given in (S-sum) is unique and is called the witness of $(f_{i_1,\dots,i_n})_{i_1,\dots,i_n\in\mathbb{N}}$. We write this witness $\langle\!\langle f_{i_1,\dots,i_n} \rangle\!\rangle_{(i_1,\dots,i_n)\in\mathbb{N}^n}$.*

The reader may be puzzled as for why we consider $\mathbb{N}^n$ indexed families in (S-fun) instead of simply $\mathbb{N}$. We need to consider these families to ensure that the iteration of S does not introduce more constraints than the summability of each component.

▶ **Example 39.** In web models, the summability structure is given by $|\mathsf{S}X| = \bigcup_{i\in\mathbb{N}} \{i\} \times |X|$ and $\mathcal{A}(\mathsf{S}X) = \left\{ (x_i)_{i\in\mathbb{N}} \in \left(\mathcal{R}_+^{|X|}\right)^{\mathbb{N}} \mid (x_i)_{i\in\mathbb{N}} \text{ is summable in } \mathcal{A}X \right\}$, with projections $\pi_i \cdot (x_i)_{i\in\mathbb{N}} = x_i$ and sum $\sigma \cdot (x_i)_{i\in\mathbb{N}} = \sum_{i\in\mathbb{N}} x_i$. Observe that $\mathsf{S}X$ sits between the product and the coproduct, as mentioned in Remark 36. We will prove in Section 7 that our web models are *representable PCM categories* (Definition 50), which will ensure *de facto* that S is a summability structure.

We do not assume in Definition 37 above that $\mathcal{L}$ is a PCM category. This is captured by (S-fun), as shown in Proposition 40 below.

▶ **Proposition 40.** *If $\mathcal{L}$ is equipped by a summability structure, then $\mathcal{L}$ is a PCM category.*

Proof. We need to prove strong distributivity. Let $\vec{f} = (f_a)_{a\in A}$ be a summable family of $\mathcal{L}(X,Y)$, and $\vec{g} = (g_a)_{a\in A}$ be a summable family of $\mathcal{L}(Y,Z)$. Let $\phi_1 : A \hookrightarrow \mathbb{N}$ and $\phi_2 : B \hookrightarrow \mathbb{N}$ be any injections (they exist because A and B are countable). Let $(f'_i)_{i\in\mathbb{N}} = \phi_1^* \vec{f}$ and $(g'_j)_{j\in\mathbb{N}} = \phi_2^* \vec{g}$. These families are summable by assumption and Proposition 9.

Let $f' \in \mathcal{L}(X, \mathsf{S}Y)$ and $g' \in \mathcal{L}(Y, \mathsf{S}Z)$ be the witnesses given by (S-sum-1). Then, by naturality of the π_i, $\pi_j \circ \pi_i \circ \mathsf{S}g' \circ f' = \pi_j \circ g' \circ \pi_i \circ f' = g'_j \circ f'_i$. Thus, by (S-sum-1), $(g'_j \circ f'_i)_{i,j\in\mathbb{N}}$ is summable with sum $\sigma \circ \sigma \circ \mathsf{S}g' \circ f' \overset{\circ 1}{=} \sigma \circ g' \circ \sigma \circ f' \overset{\circ 2}{=} \left(\sum_{j\in\mathbb{N}} g'_j\right) \circ \left(\sum_{i\in\mathbb{N}} f'_i\right) \overset{\circ 3}{=} \left(\sum_{b\in B} g_b\right) \circ \left(\sum_{a\in A} f_a\right)$ where ($\circ$1) is naturality of σ, ($\circ$2) is (S-sum-2) and ($\circ$3) is Proposition 9. To conclude the proof that $\mathcal{L}$ is a PCM category, we only need by Proposition 9 to give an injection $\psi : A \times B \to \mathbb{N}^2$ such that $(g'_j \circ f'_i)_{i,j\in\mathbb{N}} = \psi^*(h_{a,b})_{(a,b)\in A\times B}$ with $h_{a,b} = g_b \circ f_a$. Define ψ as $\psi(a,b) = (\phi_1(a), \phi_2(b))$. Then, for all $i, j \in \mathbb{N}$,

- if $(i,j) \in \mathrm{Im}(\psi)$, then $i \in \mathrm{Im}(\phi_1)$ and $j \in \mathrm{Im}(\phi_2)$, and $g'_j \circ f'_i = g_{\phi_2^{-1}(j)} \circ f_{\phi_1^{-1}(i)} = h_{\psi^{-1}(i,j)}$;
- otherwise, either $i \notin \mathrm{Im}(\phi_1)$ and $f_i = 0$, or $j \notin \mathrm{Im}(\phi_2)$ and $g_j = 0$. By (S-sum-3), 0 is a zero morphism and thus $g_j \circ f_i = 0$. ◀

As shown in [18, Sec. 3.4], every summability structure S has a canonical *bimonad* structure [40]. The proof of [18] carries to our setting, so we simply recall the result here.

▶ **Proposition 41.**
1. *For all $i \in \mathbb{N}$, there exists $\iota_i : \mathsf{Id} \Rightarrow \mathsf{S}$ characterized by $\pi_j \circ \iota_i = \delta_i^j$.*
2. *There exists a natural transformation $\theta : \mathsf{SS} \Rightarrow \mathsf{S}$ characterized by $\pi_i \circ \theta = \sum_{j=0}^{i} \pi_{i-j} \circ \pi_j$.*
3. *There exists a natural transformation $\mathsf{l} : \mathsf{S} \Rightarrow \mathsf{SS}$ characterized by $\pi_i \circ \pi_j \circ \mathsf{l} = \delta_i^j \pi_i$.*
4. *There exists a natural transformation $\mathsf{c} : \mathsf{SS} \Rightarrow \mathsf{SS}$ characterized by $\pi_i \circ \pi_j \circ \mathsf{c} = \pi_j \circ \pi_i$.*

▶ **Theorem 42** ([18, Thm. 74])**.** *$(\mathsf{S}, \iota_0, \theta)$ is a monad, $(\mathsf{S}, \sigma, \mathsf{l})$ is a comonad, $\mathsf{c} : \mathsf{SS} \Rightarrow \mathsf{SS}$ is a distributive law between the two [1, 42], and together they form a bimonad.*

By monicity of the π_i, a morphism $f \in \mathcal{L}(X, \mathsf{S}Y)$ of the Kleisli category of the monad S can be seen as a formal power series $\sum_{i\in\mathbb{N}} f_i \varepsilon^i$ over $\mathcal{L}(X,Y)$, where $f_i = \pi_i \circ f$ and ε is a formal variable. Then the composition of two power series in the Kleisli category of S is given by their Cauchy product, $\left(\sum_{j\in\mathbb{N}} g_j \varepsilon^j\right) \circ \left(\sum_{i\in\mathbb{N}} f_i \varepsilon^i\right) = \sum_{k\in\mathbb{N}} \left(\sum_{i+j=k} g_j \circ f_i\right) \varepsilon^k$. As such, the Kleisli category of S is quite reminiscent of the monoid semiring construction of [22].

Compatibility between summability and categorical structure. We detail how the PCM structure of a PCM category $\mathcal{L}$ should interact with the symmetric monoidal (closed) structure of $\mathcal{L}$, and its products, whenever they exist.

▶ **Definition 43.** *The PCM structure is compatible with the symmetric monoidal product if $\otimes$ is strongly distributive: for all indexed families $(f_a : X_1 \to Y_1)_{a\in A}$ and $(g_b : X_2 \to Y_2)_{b\in B}$, $\left(\sum_{a\in A} f_a\right) \otimes \left(\sum_{b\in B} g_b\right) \rightrightarrows \sum_{a\in A, b\in B} (f_a \otimes g_b)$.*

▶ **Definition 44.** *The PCM structure is compatible with the monoidal closure if it is compatible with the symmetric monoidal product and if for all indexed family $(f_a : X \otimes Y \to Z)_{a\in A}$, $\sum_{a\in A} \mathsf{cur}(f_a) \rightrightarrows \mathsf{cur}\left(\sum_{a\in A} f_a\right)$.*

▶ **Definition 45.** *The PCM structure is compatible with the finite products (respectively countable products) if for all finite (respectively countable) index set I and for all collection of indexed families $(f_a^i \in \mathcal{L}(X, Y_i))_{a\in A}$ where i ranges over I, $\sum_{a\in A} \langle f_a^i \rangle_{i\in I} \rightrightarrows \left\langle \sum f_a^i \right\rangle_{i\in I}$*

▶ Remark 46. These definitions are the most intuitive, but are quite redundant.
- *Monoidal product*: the left and right distributivity of $\otimes$ implies strong distributivity, by strong distributivity of the composition in PCM categories. Furthermore, left distributivity implies right distributivity by symmetry of the product, and vice versa.
- *Cartesian product*: by right distributivity, $\mathsf{p}_i \circ \sum_{a\in A} \langle f_a^i \rangle_{i\in I} \rightrightarrows \sum_{a\in A} (\mathsf{p}_i \circ \langle f_a^i \rangle_{i\in I}) = \sum_{a\in A} f_a^i$ so by uniqueness of the pairing, we always have $\sum_{a\in A} \langle f_a^i \rangle_{i\in I} \rightrightarrows \left\langle \sum f_a^i \right\rangle_{i\in I}$.
- *Monoidal closed structure*: similarly, if the PCM structure is compatible with $\otimes$, we can prove that $\sum_{a\in A} \mathsf{cur}(f_a) \rightrightarrows \mathsf{cur}\left(\sum_{a\in A} f_a\right)$.

If the PCM category structure of $\mathcal{L}$ is induced by a summability structure S, all the compatibility conditions above boil down to categorical structure on S (see proof in Appendix C).

▶ **Proposition 47.** *Assume that $\mathcal{L}$ is equipped with a summability structure S.*

1. *The PCM structure induced by S is compatible with the symmetric monoidal structure of $\mathcal{L}$ if and only if for all objects X and Y, $0 \otimes Y = 0$ and there exists*
$$\phi^1_{X,Y} = \langle\!\langle \pi_i \otimes Y \rangle\!\rangle_{i\in\mathbb{N}} : \mathsf{S}X \otimes Y \to \mathsf{S}(X \otimes Y) \text{ such that } \sigma \circ \phi^1 = (\sigma \otimes Y)$$
$$\phi^2_{X,Y} = \langle\!\langle X \otimes \pi_i \rangle\!\rangle_{i\in\mathbb{N}} : X \otimes \mathsf{S}Y \to \mathsf{S}(X \otimes Y) \text{ such that } \sigma \circ \phi^2 = (X \otimes \sigma) \tag{2}$$
2. *The PCM structure is compatible with the symmetric monoidal closed structure (whenever there is one) if and only if it is compatible with the symmetric monoidal structure, and if for all object X, $(X \multimap \pi_i)_{i\in\mathbb{N}} : (X \multimap \mathsf{S}Y) \to (X \multimap Y)$ is summable.*
3. *The PCM structure is compatible with the finite (resp. countable) products (whenever they exist) if and only if for all finite (resp. countable) index set I, $(\&_{i\in I}\, \pi_j)_{j\in\mathbb{N}}$ is summable.*

The natural transformations ϕ^1 and ϕ^2 endow S with the structure of a *commutative monad* [18]. This induces by standard results [28, 30] a *lax monoidal monad* structure $\iota_0 : 1 \to \mathsf{S}1$ and $\mathsf{L}_{X,Y} : \mathsf{S}X \otimes \mathsf{S}Y \to \mathsf{S}(X \otimes Y)$ characterized by $\pi_i \circ \mathsf{L} = \sum_{i_1+i_2=i} \pi_{i_1} \otimes \pi_{i_2}$. Similarly, S has a canonical oplax monoidal structure $\langle \mathsf{Sp}_1, \mathsf{Sp}_2 \rangle : \mathsf{S}(X \,\&\, Y) \to \mathsf{S}X \,\&\, \mathsf{S}Y$ [38, Prop. 22] with inverse $\langle\!\langle \pi_i \,\&\, \pi_i \rangle\!\rangle_{i\in\mathbb{N}} : \mathsf{S}X \,\&\, \mathsf{S}Y \Rightarrow \mathsf{S}(X \,\&\, Y)$. Thus, the compatibility between sums and products turns S into a strong monoidal monad (wrt $\&$).

Taylor expansion. Our Taylor expansion is the same as in [18].

▶ **Definition 48** ([18, Def. 112])**.** *An analytic Taylor expansion is a distributive law $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$ between the functor S and the comonad $!_$ that is also compatible with the structure on S (bimonad, monoidal structure with regard to $\otimes$ and $\&$, and projection π_0).*

6 The representable theory

We assume in this section that $\mathcal{L}$ is a symmetric monoidal closed category, with zero morphisms 0 that are also absorbing for the monoidal product : $\mathsf{id}_X \otimes 0 = 0$ and $0 \otimes \mathsf{id}_Y = 0$.

▶ **Definition 49.** *A summation object in $\mathcal{L}$ is an object $\mathbb{D}$ equipped with a family of jointly epic morphisms $(\mathsf{e}_i : 1 \to \mathbb{D})_{i\in\mathbb{N}}$ called the basis, and $\Delta : 1 \to \mathbb{D}$ called the diagonal.*

In [18], $\mathbb{D}$ is always assumed to be equal to the product $\&_{i\in\mathbb{N}}\, 1$. However, it turns out that this assumption does not give the expected notion of sum in Köthe spaces. We refer the reader to Section 7 for a generic description of this summation object in web models.

The internal hom $1 \multimap X$ can be identified with X, and then $\mathsf{ev}_{1,X} = \rho_X : X \otimes 1 \to X$ and for all $f : X \otimes 1 \to Y$, $\mathsf{cur}(f) = f \circ \rho^{-1}$. We define natural transformations $\pi_i = (\mathsf{e}_i \multimap X) : (\mathbb{D} \multimap X) \to X$ and $\sigma = (\Delta \multimap X) : (\mathbb{D} \multimap X) \to X$. More explicitely,

$$\begin{aligned} \pi_i = \quad & \mathbb{D} \multimap X \xrightarrow{\rho^{-1}} (\mathbb{D} \multimap X) \otimes 1 \xrightarrow{(\mathbb{D}\multimap X)\otimes \mathsf{e}_i} (\mathbb{D} \multimap X) \otimes \mathbb{D} \xrightarrow{\mathsf{ev}_{\mathbb{D},X}} X. \\ \sigma = \quad & \mathbb{D} \multimap X \xrightarrow{\rho^{-1}} (\mathbb{D} \multimap X) \otimes 1 \xrightarrow{(\mathbb{D}\multimap X)\otimes \Delta} (\mathbb{D} \multimap X) \otimes \mathbb{D} \xrightarrow{\mathsf{ev}_{\mathbb{D},X}} X. \end{aligned} \tag{3}$$

▶ **Definition 50.** *The category $\mathcal{L}$ is a representable PCM category if it has a summation object $\mathbb{D}$ such that $(\mathbb{D} \multimap _, (\pi_i)_{i\in\mathbb{N}}, \sigma, 0)$ is a summability structure.*

We now provide an alternative characterization of representable PCM categories, which can be obtained by "uncurrying" the definition of summability structures. By standard naturality equations on the currying, we get that for all $f : X \otimes \mathbb{D} \to Y$ and $g : X \otimes \mathbb{D}^{\otimes n} \to Y$,

$$\begin{aligned} \pi_i \circ \mathsf{cur}(f) = f \circ (X \otimes \mathsf{e}_i) \circ \rho^{-1} \qquad & \sigma \circ \mathsf{cur}(f) = f \circ (X \otimes \Delta) \circ \rho^{-1} \\ \pi_{i_1} \circ \cdots \circ \pi_{i_n} \circ \mathsf{cur}^n(g) = g \circ (X \otimes \mathsf{e}_{i_n} \otimes \cdots \otimes \mathsf{e}_{i_1}) \circ (\rho^n)^{-1} & \\ \sigma \circ \cdots \circ \sigma \circ \mathsf{cur}^n(g) = g \circ (X \otimes \Delta \otimes \cdots \circ \Delta) \circ (\rho^n)^{-1} & \end{aligned} \tag{4}$$

▶ **Lemma 51.** *For all object $\mathbb{D}$ and every family of morphisms $(\mathsf{e}_i : 1 \to \mathbb{D})_{i\in\mathbb{N}}$, define π_i as in Equation (3). Then, the following assertions are equivalent:*

1. *the morphisms e_i are jointly epic;*
2. *for all object X, the morphisms $X \otimes \mathsf{e}_i : X \otimes 1 \to X \otimes \mathbb{D}$ are jointly epic;*
3. *the morphisms π_i are jointly monic.*

Proof. (1) ⇒ (2). Let $f, g : X \otimes \mathbb{D} \to Y$ be such that $f \circ (X \otimes \mathsf{e}_i) = g \circ (X \otimes \mathsf{e}_i)$ for all $i \in \mathbb{N}$. By naturality of γ, we get $f\circ\gamma\circ(\mathsf{e}_i\otimes X) = g\circ\gamma\circ(\mathsf{e}_i\otimes X)$ and hence $\mathsf{cur}(f\circ\gamma)\circ\mathsf{e}_i = \mathsf{cur}(g\circ\gamma)\circ\mathsf{e}_i$ so that $\mathsf{cur}(f \circ \gamma) = \mathsf{cur}(g \circ \gamma)$ and hence $f = g$. So the $X \otimes \mathsf{e}_i$ are jointly epic. Conversely, (2) ⇒ (1), taking $X = 1$. Finally, (2) ⇔ (3) by Equation (4) and bijectivity of cur. ◀

Thus, the only condition that does not hold de facto when there is a summation object is (S-sum). This condition can be uncurried by Equation (4) as follows.

▶ **Proposition 52.** *The category $\mathcal{L}$ is a representable PCM category if and only if it has a summation object $\mathbb{D}$ such that for all objects X, Y, $\mathcal{L}(X, Y)$ has a PCM structure in which:*

1. *A family $(f_{i_1,\dots,i_n})_{i_n,\dots,i_n\in\mathbb{N}}$ of morphisms $X \to Y$ is summable if and only if there exists $h : X \otimes \mathbb{D}^{\otimes n} \to Y$ such that $h \circ (X \otimes \mathsf{e}_{i_n} \otimes \cdots \otimes \mathsf{e}_{i_1}) \circ (\rho^n_X)^{-1} = h_{i_1,\dots,i_n}$.*
2. *Then, $\sum_{i_1,\dots,i_n\in\mathbb{N}} h_{i_1,\dots,i_n} = h \circ (X \otimes \Delta \otimes \cdots \otimes \Delta) \circ (\rho^n_X)^{-1}$.*
3. *The sum over the empty family is equal to the zero morphism 0*

It immediately follows from (Unary) and from the characterization above that there exists for all $i \in \mathbb{N}$ a projection $\mathsf{pr}_i : \mathbb{D} \to 1$ such that $\mathsf{pr}_i \circ \mathsf{e}_j = \delta^j_i$ and $\mathsf{pr}_i \circ \Delta = \mathsf{id}_1$. As such, $\mathbb{D}$ is quite similar to a biproduct, except that the pairing and the copairing do not always exist.

▶ **Theorem 53.** *In every representable PCM category, the sum is compatible with the monoidal product, the closure, and the cartesian product (whenever it exists).*

To prove Theorem 53, we check that the summability structure $\mathbb{D} \multimap _$ satisfies the conditions of Proposition 47 (see Appendix D).

We now adapt to our PCM setting one of the main results of [18]: there is a bijection between Taylor expansions $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$ and *analytic coalgebra* structures on $\mathbb{D}$.

The object $\mathbb{D}$ can be equipped with a comonoid structure, given by $\mathsf{pr}_0 : \mathbb{D} \to 1$ and $\widetilde{\theta} : \mathbb{D} \to \mathbb{D} \otimes \mathbb{D}$ characterized by $\widetilde{\theta} \circ \mathsf{e}_n = \left(\sum_{i+j=n}(\mathsf{e}_i \otimes \mathsf{e}_j)\right) \rho_1^{-1}$. The morphism $\widetilde{\theta}$ exists because $(\mathsf{e}_i \otimes \mathsf{e}_j)_{i,j\in\mathbb{N}}$ is summable by Proposition 52, so $\left(\sum_{i+j=n} \mathsf{e}_i \otimes \mathsf{e}_j\right)_{n\in\mathbb{N}}$ is summable by (WPA). The object $\mathbb{D}$ can also be equipped with a monoid structure, given by $\Delta : 1 \to \mathbb{D}$ and $\widetilde{1} : \mathbb{D} \otimes \mathbb{D} \to \mathbb{D}$ characterized by $\widetilde{1} \circ (\mathsf{e}_i \otimes \mathsf{e}_j) = \delta^j_i \circ \mathsf{e}_i$. The morphism $\widetilde{1}$ exists because the family $(\mathsf{e}_i)_{i\in\mathbb{N}}$ is summable by Proposition 52, so $(\delta^j_i \mathsf{e}_i)_{i,j\in\mathbb{N}}$ is summable by Proposition 8.

▶ **Theorem 54** ([18, Thm. 212])**.** *The tuple $(\mathbb{D}, \Delta, \widetilde{1}, \mathsf{pr}_0, \widetilde{\theta})$ is a bicommutative bimonoid.*

The proof of [18] also works for our PCMs. The bimonoid structure on $\mathbb{D}$ is related to the bimonad structure of S through the *mate isomorphism* [24] (see Figure 2). The bimonad S can be seen as a kind of writer/reader bimonad associated to the bimonoid $\mathbb{D}$.

▶ **Definition 55** ([18, Def. 218])**.** *An analytic coalgebra is a $!$-coalgebra structure $\widetilde{\partial} : \mathbb{D} \to !\mathbb{D}$ such that the bimonoid structure of $\mathbb{D}$ and $\mathsf{e}_0 : 1 \to \mathbb{D}$ are $!$-coalgebra morphisms.*

▶ **Theorem 56** ([18, Cor. 222])**.** *There is a bijection between analytic Taylor expansions $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$ for $\mathsf{S} = \mathbb{D} \multimap _$ and analytic coalgebras.*

The proof of [18] also carries directly to our setting (it does not use any assumption on summability, only a purely categorical construction involving the mate isomorphism).

Bimonoid $\mathbb{D}$		Bimonad $_ \otimes \mathbb{D}$		Bimonad $\mathbb{D} \multimap _$
Projection $\mathsf{pr}_i \in \mathcal{L}(\mathbb{D}, 1)$		$\rho \circ (X \otimes \mathsf{pr}_i) \in \mathcal{L}(X \otimes \mathbb{D}, X)$		Injection ι_i
Basis $\mathsf{e}_i \in \mathcal{L}(1, \mathbb{D})$		$(X \otimes \mathsf{e}_i) \circ \rho^{-1} \in \mathcal{L}(X, X \otimes \mathbb{D})$		Projection π_i
Monoid unit Δ		Monad unit $(X \otimes \Delta) \circ \rho^{-1}$		Comonad unit σ
Monoid multiplication $\widetilde{\mathsf{l}}$		Monad sum $(X \otimes \widetilde{\mathsf{l}}) \circ \alpha$		Comonad sum l
Comonoid unit p_0	$\Longleftrightarrow$	Comonad unit $\rho \circ (X \otimes \mathsf{p}_0)$	$\overset{\text{mates}}{\Longleftrightarrow}$	Monad unit ι_0
Comonoid multiplication $\widetilde{\theta}$		Comonad sum $\alpha \circ (X \otimes \widetilde{\theta})$		Monad sum θ
Commutativity γ		Distributive law $X \otimes \gamma$		Distributive law c
Analytic coalgebra $\widetilde{\partial} \in \mathcal{L}(\mathbb{D}, !\mathbb{D})$		Distributive law $(!_) \otimes \mathbb{D} \Rightarrow !(_ \otimes \mathbb{D})$		Distributive law $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$

Figure 2 Bimonoid and bimonad relations.

7 Taylor expansion in web models

We prove in this section that our web models are representable PCM categories, and feature a Taylor expansion. For all set A and all PCR $\mathcal{R}$, define $\Delta_A \in \mathcal{R}^A$ by $(\Delta_A)_a = 1$. We then define $l^1(A) \subseteq \mathcal{R}^A$ and $l^\infty(A) \subseteq \mathcal{R}^A$ by $l^1(A) = \{\Delta_A\}^\perp$ and $l^\infty(A) = \{\Delta_A\}^{\perp\perp}$. In particular, $(x_a)_{a\in A} \in l^1(A)$ if and only if $(x_a)_{a\in A}$ is summable and $\sum_{a\in A} x_a \in \mathfrak{B}$. Then, we set $\mathbb{D} = (\mathbb{N}, l^\infty(\mathbb{N}))$.

▶ **Example 57.**

- In probabilistic coherence spaces [11], $l^1(A)$ is the set of all sub probability distributions on A, and $l^\infty(A)$ is the set $[0,1]^A$.
- In finiteness spaces [13], $l^1(A)$ is the set of all A-indexed families with finite support, and $l^\infty(A)$ is the set of all A-indexed families.
- In Köthe spaces [12], $l^1(A)$ is the set of all absolutely convergent families, and $l^\infty(A)$ is the set of all bounded families.

More often than not, $l^\infty(A) = \mathcal{A}\left(\&_{a\in A}\, 1\right)$ so $\mathbb{D} = \&_{i\in\mathbb{N}}\, 1$, as assumed in [18]. However, this is not true in Köthe spaces in which $\mathcal{A}\left(\&_{a\in A}\, 1\right)$ is the set of all A-indexed families.

The object $\mathbb{D}$ is a summation object, with diagonal $\Delta = \Delta_{\mathbb{N}}$ and basis $\mathsf{e}_i \in \mathcal{R}^{|\mathbb{D}|}$ defined by $(\mathsf{e}_a)_{a'} = \delta_a^{a'}$ (they belong to $\mathcal{A}\mathbb{D}$ by downward closure, since $\mathsf{e}_a \leq \Delta$). We now prove that the summability in $\mathbf{Web}_\mathcal{R}$ described in Section 4 is representable.

▶ **Lemma 58.** *For all $k \in \mathbb{N}$, $\mathcal{A}\left(\mathbb{D}^{\otimes k}\right) = l^\infty(\mathbb{N}^k)$.*

Proof. $\mathcal{A}\left(\mathbb{D}^{\otimes k}\right) = \left(\{\Delta\}^{\perp\perp} \otimes \cdots \otimes \{\Delta\}^{\perp\perp}\right)^{\perp\perp} \overset{\circ 1}{=} (\{\Delta\} \otimes \cdots \otimes \{\Delta\})^{\perp\perp} = \{\Delta_{\mathbb{N}^k}\}^{\perp\perp} = l^\infty(\mathbb{N}^k)$ where $(\circ 1)$ is obtained by iterating Corollary 24. ◀

▶ **Theorem 59.** *The sum in $\mathbf{Web}_{\mathcal{R}_+}$ and $\mathbf{Web}_\mathcal{R}$ is representable.*

Proof. It suffices to do the proof for $\mathbf{Web}_\mathcal{R}$ ($\mathbf{Web}_{\mathcal{R}_+}$ is a special instance of $\mathbf{Web}_\mathcal{R}$ in which $|.|$ is the identity function). We prove that $\mathbb{D}$ satisfies the conditions of Proposition 52. Let $(f(i_1,\ldots,i_k))_{i_1,\ldots,i_k\in\mathbb{N}}$ be a family of $\mathcal{R}\langle X \multimap Y\rangle$. Define $h \in \mathcal{R}^{|X\otimes\mathbb{D}^{\otimes k}\multimap Y|}$ by $h_{a,i_k,\ldots,i_1,b} = f(i_1,\ldots,i_k)_{a,b}$ so that $f(i_1,\ldots,i_k) = h \cdot (X \otimes \mathsf{e}_{i_k} \otimes \cdots \otimes \mathsf{e}_{i_1}) \cdot (\rho^k)^{-1}$ and $\sum_{i_1,\ldots,i_k\in\mathbb{N}} f(i_1,\ldots,i_k) \leftrightarrows h \cdot (X \otimes \Delta \otimes \cdots \otimes \Delta) \cdot (\rho^k)^{-1} \leftrightarrows h \cdot (X \otimes \Delta_{\mathbb{N}^k}) \cdot (\rho^k)^{-1}$. It suffices to prove

that $(f(i_1,\dots,i_k))_{i_1,\dots,i_k\in\mathbb{N}}$ is summable in $\mathcal{R}\langle X \multimap Y\rangle$ if and only if $h \in \mathcal{R}\langle X \otimes \mathbb{D}^{\otimes k} \multimap Y\rangle$ to conclude. We proceed by the following chain of equivalences.

$$
\begin{aligned}
&(f_{i_1,\dots,i_k})_{i_1,\dots,i_k\in\mathbb{N}} \text{ is summable in } \mathcal{R}\langle X \multimap Y\rangle \\
&\iff (\mathsf{abs}\,(f)_{i_1,\dots,i_k})_{i_1,\dots,i_k\in\mathbb{N}} \text{ is summable in } \mathcal{A}(X \multimap Y) \\
&\overset{\circ 1}{\iff} \forall x \in \mathcal{A}X, \forall y' \in \mathcal{A}Y^{\perp}, \left\langle \sum_{i_1,\dots,i_k\in\mathbb{N}} \mathsf{abs}\,(f)_{i_1,\dots,i_k} \,\middle|\, x \otimes y' \right\rangle \text{is defined and in } \mathfrak{B} \\
&\overset{\circ 2}{\iff} \forall x \in \mathcal{A}X, \forall y' \in \mathcal{A}Y^{\perp}, \langle \mathsf{abs}\,(h) \mid x \otimes \Delta_{\mathbb{N}^k} \otimes y' \rangle \text{ is defined and in } \mathfrak{B} \\
&\overset{\circ 3}{\iff} \mathsf{abs}\,(h) \in \mathcal{A}(X \otimes \mathbb{D}^{\otimes k} \multimap Y) \iff h \in \mathcal{R}\langle X \otimes \mathbb{D}^{\otimes k} \multimap Y\rangle
\end{aligned}
$$

where (°1) is the predual characterization of summability given in Proposition 33, (°2) holds by the covering principle of Lemma 21 ($\{\Delta_{\mathbb{N}}\}$ is a covering), and (°3) holds by the predual characterization of Theorem 23, using that $\mathcal{A}\left(\mathbb{D}^{\otimes k}\right) = \{\Delta_{\mathbb{N}^k}\}^{\perp\perp}$ by Lemma 58. ◀

Finally, the analytic coalgebra structure on $\mathbb{D}$ is given by $\widetilde{\partial} \in \mathcal{R}^{|\mathbb{D}\multimap !\mathbb{D}|}$ with $\widetilde{\partial}_{i,[i_1,\dots,i_n]} = \delta_i^{i_1+\dots+i_n}$, so that $(\widetilde{\partial}\cdot x)_{[i_1,\dots,i_n]} = x_{i_1+\dots+i_n}$. Independently of any specificities of the model, we always have that $\mathsf{abs}\left(\widetilde{\partial}\right)\cdot\mathsf{abs}\,(\Delta) = \mathsf{abs}\left(\Delta^!\right) \in \mathcal{A}(!\mathbb{D})$ so by Theorem 23, $\widetilde{\partial} \in \mathcal{R}\langle \mathbb{D} \multimap !\mathbb{D}\rangle$. Thus, this analytic coalgebra generalizes to our generic web models the analytic coalgebra of the weighted relational model [18]. This analytic coalgebra induces a Taylor expansion $\partial : !\mathsf{S} \Rightarrow \mathsf{S}!$ that in turns induces a functor T on $\mathcal{L}_!$. For all $s \in \mathcal{R}\langle !X \multimap Y\rangle$, $\mathsf{T}s \in \mathcal{R}^{|!\mathsf{S}X\multimap \mathsf{S}Y|}$ is defined as in the examples of [18] by

$$
(\mathsf{T}s)_{[(i_1,a_1),\dots,(i_k,a_k)],(j,b)} = \delta^j_{i_1+\dots+i_k} \frac{[a_1,\dots,a_k]!}{[(i_1,a_1),\dots,(i_k,a_k)]!} s_{[a_1,\dots,a_k],b}. \tag{5}
$$

We know *by construction* that $\mathsf{T}s$ belongs to $\mathcal{R}\langle !\mathsf{S}X \multimap \mathsf{S}Y\rangle$, for *all* our web models. It is proved in [18] that this functor corresponds formally to the intuition given in Equation (1), using the differential linear logical structure of the weighted relational model [2].

8 Conclusion and future work

In Part I, we have built a generic construction of web models, based on strong and absolute PCMs. It unifies various models of the literature. In Part II, we have described a theory of Taylor expansion in categories enriched over PCMs. Web models given by our generic construction feature such Taylor expansion as proved in Section 7. Thus, Taylor expansion is compatible with the computational properties described by partial sums, such as determinism, finite or countable non-determinism, randomness, and absolute convergence.

Future work. We conjecture that the exponential of our web models is always Lafont (see Remark 25). Our theory of coherent Taylor expansion captures a wide class of web models, but there are no known web-free examples. We conjecture that the analytic version [26] of the category of convenient vector spaces [4] is not a representable PCM category but features a coherent Taylor expansion. Finally, although we described our web models with a strong emphasis on the *algebraic* point of view of partial sums, it should be possible to adopt a topological viewpoint on absolute PCMs, as it is done in Köthe spaces [12] and finiteness spaces [13].

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A Proofs on web based models

Proof of Lemma 20. Observe that

$$\langle s \cdot x \mid y \rangle = \sum_{b\in|Y|} \Big(\sum_{a\in|X|} s_{a,b}x_a \Big) y_b \overset{\circ 1}{=\!\!=} \sum_{b\in|Y|} \sum_{a\in|X|} (s_{a,b}x_a y_b) \overset{\circ 2}{=\!\!=} \sum_{a\in|X|,b\in|Y|} s_{a,b}x_a y_b = \langle s \mid x \otimes y \rangle$$

where ($\circ$1) is by distributivity of multiplication over sums, and ($\circ$2) is by (PA). Furthermore, if $s \cdot x$ is defined, then ($\circ$1) turns into an equivalence $=\!\!=$ (but not in general, consider $y_b = 0$). The second item is proved in the same way. ◀

Proof of Lemma 21. Let $b_0 \in |Y|$. We prove that $(s \cdot x)_{b_0}$ is defined. By covering, there exists $y \in F$ such that y_{b_0} is invertible. By assumption, $(s_{a,b}x_a y_b)_{a\in|X|,b\in|Y|}$ is summable, so by Lemma 3 $(s_{a,b_0}x_a y_{b_0})_{a\in|X|}$ is summable. But then $y_{b_0}^{-1}\big(\sum_{a\in|X|} s_{a,b_0}x_a y_{b_0}\big) =\!\!= \sum_{a\in|X|} s_{a,b_0}x_a y_{b_0} y_{b_0}^{-1} = \sum_{a\in|X|} s_{a,b_0}x_a$ so $(s_{a,b_0}x_a)_{a\in|X|}$ is summable. ◀

Proof of Proposition 22. We first prove (1) $\iff$ (2). By Remark 18,

$$s \in \mathcal{A}(X \multimap Y) \iff \forall x \in \mathcal{A}X, \forall y' \in \mathcal{A}Y, \langle s \mid x \otimes y' \rangle \text{ is defined and in } B.$$
$$s^\perp \in \mathcal{A}(Y^\perp \multimap X^\perp) \iff \forall x \in \mathcal{A}X, \forall y' \in \mathcal{A}Y, \langle s^\perp \mid y' \otimes x \rangle \text{ is defined and in } B.$$

We can easily check that $\langle s \mid x \otimes y' \rangle =\!\!= \langle s^\perp \mid y' \otimes x \rangle$, which concludes the proof of equivalence. Next, we prove (1) $\Rightarrow$ (3). Let $s \in \mathcal{A}(X \multimap Y)$ and $x \in \mathcal{A}X$. First, observe that $s \cdot x$ is well-defined by covering principle (Lemma 21), since $\mathcal{A}Y^\perp$ is a covering. Then, for all

$y' \in Y^\perp$, $\langle s \cdot x \mid y' \rangle \rightleftarrows \langle s \mid x \otimes y' \rangle$ by Lemma 20. Since $s \perp x \otimes y'$ by assumption, we get that $s \cdot x \perp y'$. It concludes the proof of (3). The proof of the reverse implication (3) $\Rightarrow$ (1) relies on Lemma 20 in a similar same way. Finally, (2) $\iff$ (4) holds using (1) $\iff$ (3) on $s^\perp$. ◀

Proof of Theorem 19. Let $s \in \mathcal{A}(X \multimap Y)$ and $t \in \mathcal{A}(Y \multimap Z)$. First, let us prove that $t \cdot s$ is well-defined. By covering, for all $a_0 \in |X|$ there exists $x \in \mathcal{A}X$ such that x_{a_0} is invertible. Then, $(t \cdot (s \cdot x))$ is defined by iterating Proposition 22 twice. But

$$(t \cdot (s \cdot x))_c = \sum_{b \in |Y|} t_{b,c} \Big(\sum_{a \in |X|} s_{a,b} x_a \Big) \rightrightarrows \sum_{b \in |Y|} \sum_{a \in |X|} t_{b,c} s_{a,b} x_a \overset{\text{(PA)}}{\rightrightarrows} \sum_{a \in |X|, b \in |Y|} s_{a,b} t_{b,c} x_a.$$

It follows from (WPA) that $(s_{a_0,b} t_{b,x} x_{a_0})_{b \in |Y|}$ is summable, so by invertibility of x_{a_0}, $(s_{a_0,b} t_{b,c})_{b \in |Y|}$ is summable. That is, $t \cdot s$ is well-defined. We now prove that it belongs to $\mathcal{A}(X \multimap Z)$ using Proposition 22. Let $x \in \mathcal{A}X$. Then,

$$((t \cdot s) \cdot x)_c = \sum_{a \in |X|} \Big(\sum_{b \in |Y|} s_{a,b} t_{b,c} \Big) x_a \overset{\circ 1}{\rightrightarrows} \sum_{a \in |X|} \Big(\sum_{b \in |Y|} s_{a,b} t_{b,c} x_a \Big).$$

$$(t \cdot (s \cdot x))_c = \sum_{b \in |Y|} t_{b,c} \Big(\sum_{a \in |X|} s_{a,b} x_a \Big) \overset{\circ 2}{\rightrightarrows} \sum_{b \in |Y|} \Big(\sum_{a \in |X|} s_{a,b} t_{b,c} x_a \Big).$$

Be aware: (°1) only holds because we already know that $t \cdot s$ is defined, and (°2) only holds because we already know that $s \cdot x$ is defined. Then, by (PA), $(t \cdot s) \cdot x \rightrightarrows t \cdot (s \cdot x)$. We conclude that $t \cdot s$ is well-defined and belongs to $\mathcal{A}(X \multimap Z)$ by iterating Proposition 22 twice. ◀

Proof of Theorem 23. First, observe that (1) $\Rightarrow$ (2) is trivial.

(2) $\Rightarrow$ (3). Let $s \in (P \otimes Q)^\perp$. Then for all $x \in P$, $s \cdot x$ is well-defined by covering principle (Lemma 21). Furthermore, for all $y' \in Q$, $\langle s \mid x \otimes y' \rangle \rightleftarrows \langle s \cdot x \mid {}'y' \rangle$ by Lemma 20. But $s \perp x \otimes y'$ so we deduce that $s \cdot x \perp y'$. Thus, $s \cdot x \in Q^\perp = \mathcal{A}Y$.

(3) $\Rightarrow$ (2). Assume that s satisfies (3). Then we prove that $s \in (P \otimes \mathcal{A}Y^\perp)^\perp \subseteq (P \otimes Q)^\perp$. Let $x \in P$ and $y' \in (\mathcal{A}Y)^\perp$. Then by assumption $s \cdot x$ is defined so by Lemma 20 $\langle s \mid x \otimes y' \rangle \rightleftarrows \langle s \cdot x \mid y' \rangle$ and we have that $s \perp x \otimes y'$. Thus, $s \in (P \otimes \mathcal{A}Y^\perp)^\perp$.

(2) $\iff$ (4). The equivalence holds by a similar argument, since $\langle s \mid x \otimes y' \rangle \rightleftarrows \langle s^\perp \mid y' \otimes x \rangle$.

(3) $\wedge$ (4) $\Rightarrow$ (1). Assume that s satisfies (3) and (4). First, we know from the proof of (3) $\Rightarrow$ (2) that $s \in (P \otimes \mathcal{A}Y^\perp)^\perp$. Second, by the implication (2) $\Rightarrow$ (4) applied on P and $\mathcal{A}Y^\perp$, we know that for all $y' \in \mathcal{A}Y^\perp$, $s^\perp \cdot y'$ is defined and belongs to $\mathcal{A}X^\perp$. So by the characterization of Proposition 22, we conclude that $s \in \mathcal{A}(X \multimap Y)$. ◀

B Summability structures, products and coproducts

As stated in Proposition 41, for all $i \in \mathbb{N}$ and all object X there exists a morphism $\iota_i \in \mathcal{L}(X, \mathsf{S}X)$ such that $\pi_j \circ \iota_i = \delta_i^j$ and $\sigma \circ \iota_i = \mathsf{id}$. We prove that there is a *factorization* $\bigoplus_{i=0}^n X \xrightarrow{[\iota_i]_{i \in \mathbb{N}}} \mathsf{S}X \xrightarrow{\langle \pi_i \rangle_{i \in \mathbb{N}}} \prod_{i=0}^n X$.

Observe that $\langle \pi_i \rangle_{i \in \mathbb{N}}$ is a mono, by joint monicity of the π_i (S-proj) and uniqueness of the pairing. We now prove that $[\iota_i]_{i \in \mathbb{N}} : \bigoplus_{i=0}^n X \to \mathsf{S}X$ is an epi. For a similar reason, it suffices to prove that the ι_i are jointly epic. This is done in Proposition 60 and Corollary 61 below.

▶ **Proposition 60.** *The family $(\iota_i \circ \pi_i)_{i \in \mathbb{N}}$ of $\mathcal{L}(\mathsf{S}X, \mathsf{S}X)$ is summable, with sum $\mathsf{id}_{\mathsf{S}X}$.*

Proof. Observe that $\pi_j \circ \iota_i \circ \pi_i = \delta_i^j\ \pi_i$. It follows from the summability of $(\pi_i)_{i\in\mathbb{N}}$ and Proposition 8 that $(\pi_j \circ \iota_i \circ \pi_i)_{i,j\in\mathbb{N}}$ is summable. But then, $\pi_j \circ \pi_i \circ \langle\!\langle \pi_j \circ \iota_i \circ \pi_i \rangle\!\rangle_{(j,i)\in\mathbb{N}^2} = \pi_j \circ \iota_i \circ \pi_i$ so by joint monicity of the π_j, $\pi_i \langle\!\langle \pi_j \circ \iota_i \circ \pi_i \rangle\!\rangle_{(j,i)\in\mathbb{N}^2} = \iota_i \circ \pi_i$ and $(\iota_i \circ \pi_i)_{i\in\mathbb{N}}$ is summable by (S-sum). Then $\pi_k \circ \left(\sum_{i\in\mathbb{N}} \iota_i \circ \pi_i\right) \rightrightarrows \sum_{i\in\mathbb{N}} (\pi_k \circ \iota_i \circ \pi_i) \rightrightarrows \sum_{i\in\mathbb{N}} (\delta_i^k\ \pi_i) = \pi_k$ by (Unary), so by joint monicity of the π_k, $\sum_{i\in\mathbb{N}} \iota_i \circ \pi_i = \mathsf{id}_{\mathsf{S}X}$. ◀

▶ **Corollary 61.** *The ι_i are jointly epic: $\forall f, g : \mathsf{S}X \to Y$, $(\forall i \in \mathbb{N}, f \circ \iota_i = g \circ \iota_i) \Rightarrow f = g$.*

Proof. $f = f \circ \left(\sum_{i\in\mathbb{N}} \iota_i \circ \pi_i\right) \rightrightarrows \sum_{i\in\mathbb{N}} f \circ \iota_i \circ \pi_i = \sum_{i\in\mathbb{N}} g \circ \iota_i \circ \pi_i \rightrightarrows g \circ \left(\sum_{i\in\mathbb{N}} \iota_i \circ \pi_i\right) = g$. ◀

C Proof of Proposition 47

Proof of (1). The forward implication is trivial, because Equation (2) is a particular instance of Definition 43. The proof of the reverse direction is very similar to the proof of Proposition 40. Let $g : X_2 \to Y_2$ be a morphism, and $(f_i)_{a\in A}$ be a summable family of morphisms in $X_1 \to X_2$. Let $\psi : A \hookrightarrow \mathbb{N}$ be an injection. Let $(f'_i)_{i\in\mathbb{N}} = \psi^*(f_a)_{a\in A}$. This family is summable by Proposition 9. Let $h = \phi^1_{Y_1,Y_2} \circ \left(\langle\!\langle f'_i \rangle\!\rangle_{i\in\mathbb{N}} \otimes g\right) : X_1 \otimes Y_2 \to \mathsf{S}(Y_1 \otimes Y_2)$.

Then we can easily check (by definition of ϕ^1) that $\pi_i \circ h = f'_i \otimes g$ and $\sigma \circ h = \left(\sum_{i\in\mathbb{N}} f'_i\right) \otimes g$, so by (S-sum) $(f'_i)_{i\in\mathbb{N}}$ is summable with sum $\left(\sum_{a\in A} f_a\right) \otimes g$. We conclude that $(f_a)_{a\in A}$ is summable with the same sum, by reindexing (Proposition 9), and using that $0 \otimes g = 0$. So $\otimes$ is left distributive. We conclude that $\otimes$ is strongly distributive by Remark 46. ◀

Proof of (2). For the forward direction, recall that $X \multimap \pi_i : (X \multimap \mathsf{S}Y) \to (X \multimap Y)$ is defined as $\mathsf{cur}\left(\mathsf{ev} \circ ((X \multimap Y) \otimes \pi_i)\right)$. But $\mathsf{ev} \circ ((X \multimap Y) \otimes \pi_i)$ is summable, by summability of the π_i, right distributivity (of $\otimes$) and right distributivity (of composition). So the $X \multimap \pi_i$ are summable. Conversely, it suffices by Remark 46 to prove that for all summable family $(f_a)_{a\in A}$ of $\mathcal{L}(X \otimes Y, Z)$, $(\mathsf{cur}(f_a))_{a\in A}$ is summable. Let $\psi : A \hookrightarrow \mathbb{N}$ be an injection. Let $(f'_i)_{i\in\mathbb{N}} = \psi^*(f_a)_{a\in A}$. This family is summable by Proposition 9. Let

$$h = \langle\!\langle Y \multimap \pi_i \rangle\!\rangle_{i\in\mathbb{N}} \circ \mathsf{cur}\left(\langle\!\langle f'_i \rangle\!\rangle_{i\in\mathbb{N}}\right) : X \to \mathsf{S}(Y \multimap Z).$$

Then we can check that $\pi_i \circ h = \mathsf{cur}(f'_i)$, so $(\mathsf{cur}(f'_i))_{i\in\mathbb{N}}$ is summable. But $(\mathsf{cur}(f'_i))_{i\in\mathbb{N}} = \psi^*(\mathsf{cur}(f_a))_{a\in A}$ (because $\mathsf{cur}(0) = 0$) so we conclude that $(\mathsf{cur}(f_a))_{a\in A}$ is summable. ◀

Proof of (3). For the forward direction, observe that $\&_{i\in I}\, \pi_j = \langle \pi_j \circ \mathsf{p}_i \rangle_{i\in I}$ is a pairing of summable morphisms. The proof of the reverse direction is similar to the proof of (2). ◀

D Results on representable summability

Let $\mathcal{L}$ be a representable PCM category. We prove in this section that $\mathcal{L}$ satisfies Theorem 53.

Compatibility with the monoidal product. We use the characterization of Proposition 47. We already know by assumption that 0 is absorbing for the monoidal product. Let us prove that $(\pi_i \otimes X_2)_{i\in\mathbb{N}}$ is a summable family of $\mathcal{L}((\mathbb{D} \multimap X_1) \otimes X_2, X_1 \otimes X_2)$, with sum $\sigma \otimes X_1$. Let $f = (\mathbb{D} \multimap X_1) \otimes X_2 \otimes \mathbb{D} \xrightarrow{(\mathbb{D}\multimap X_1)\otimes\gamma} (\mathbb{D} \multimap X_1) \otimes \mathbb{D} \otimes X_2 \xrightarrow{\mathsf{ev}\otimes X_2} X_1 \otimes X_2$. Then,

$$f \circ (\mathsf{id}_{\mathbb{D}\multimap X_1} \otimes X_2 \otimes \mathsf{e}_i) \circ \rho^{-1}_{\mathbb{D}\multimap X_1 \otimes X_2} \overset{\circ 1}{=} (\mathsf{ev}\otimes X_2) \circ (\mathsf{id}_{\mathbb{D}\multimap X_1} \otimes \mathsf{e}_i \otimes X_2) \circ (\rho^{-1}_{\mathbb{D}\multimap X_1} \otimes X_2) \overset{\circ 2}{=} \pi_i \otimes X_2$$

where ($\circ$1) is obtained by standard computations on symmetric monoidal categories, and ($\circ$2) is an immediate consequence of Equation (3). So by Proposition 52, $(\pi_i \otimes X_2)_{i\in\mathbb{N}}$ is summable with sum $f \circ ((\mathbb{D} \multimap X_1) \otimes X_2 \otimes \Delta) \circ \rho^{-1}_{\mathbb{D}\multimap X_1 \otimes X_2} = \sigma \otimes X_2$ by a similar computation. The summability of $(X_1 \otimes \pi_i)_{i\in\mathbb{N}}$ follows by symmetry of the monoidal product, see Remark 46. ◀

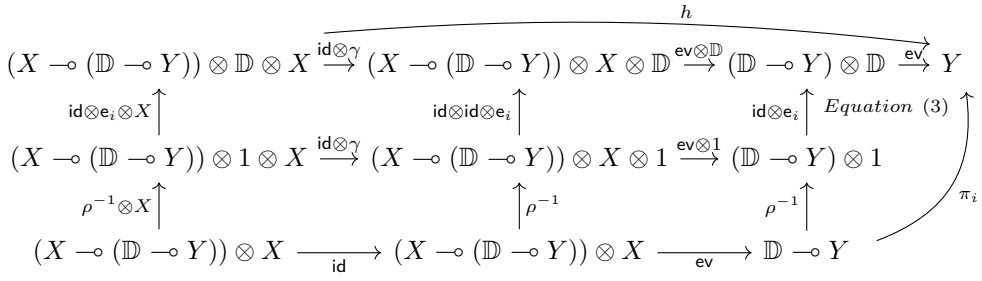
Compatibility with the symmetric monoidal closed structure. By Proposition 47 is suffices to prove that $(X \multimap \pi_i)_{i\in\mathbb{N}}$ is summable in $\mathcal{L}(X \multimap (\mathbb{D} \multimap Y), X \multimap Y)$. Define

$$h = \ (X \multimap (\mathbb{D} \multimap Y)) \otimes \mathbb{D} \otimes X \xrightarrow{\mathsf{id}\otimes\gamma} (X \multimap (\mathbb{D} \multimap Y)) \otimes X \otimes \mathbb{D} \xrightarrow{\mathsf{ev}\otimes\mathbb{D}} (\mathbb{D} \multimap Y) \otimes \mathbb{D} \xrightarrow{\mathsf{ev}} Y$$

It suffices to prove that $\mathsf{cur}(h) : (X \multimap (\mathbb{D} \multimap Y)) \otimes \mathbb{D} \to (X \multimap Y)$ is a witness for $(X \multimap \pi_i)_{i\in\mathbb{N}}$ in the sense of Proposition 52 to conclude.

$$\begin{aligned}\mathsf{cur}(h)\circ\big(\mathsf{id}_{X\multimap(\mathbb{D}\multimap Y)}\otimes\mathsf{e}_i\big)\circ\rho^{-1}_{X\multimap(\mathbb{D}\multimap Y)} &\overset{\circ 1}{=} \mathsf{cur}\big(h\circ(\mathsf{id}_{X\multimap(\mathbb{D}\multimap Y)}\otimes\mathsf{e}_i\otimes X)\circ(\rho^{-1}_{X\multimap(\mathbb{D}\multimap Y)}\otimes X)\big)\\ &\overset{\circ 2}{=} \mathsf{cur}(\pi_i\circ\mathsf{ev}_{X,\mathbb{D}\multimap Y}) = X\multimap\pi_i\end{aligned}$$

where ($\circ$1) is by naturality of cur and ($\circ$2) is the diagram chase below.



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Compatibility with the products. Let I be a set such that I-indexed products exist. By Proposition 47, it suffices to prove that $(\&_{i\in I}\,\pi_j)_{i\in j}$ is summable, which amounts by Proposition 52 to the construction of a witness $\big(\&_{i\in I}(\mathbb{D}\multimap X_i)\big)\otimes\mathbb{D}\to\&_{i\in I}\,X_i$. The morphism $f = \langle\mathsf{ev}\circ(\mathsf{p}_i\otimes\mathbb{D})\rangle_{i\in I}$ works by ($\circ$1) functoriality and naturality and ($\circ$2) Equation (3).

$$\begin{aligned}f\circ(\&_{i\in I}\,X_i\otimes\mathsf{e}_j)\circ\rho^{-1} &= \big\langle\mathsf{ev}\circ(\mathsf{p}_i\otimes\mathsf{e}_j)\circ\rho^{-1}\big\rangle_{i\in I}\\ &\overset{\circ 1}{=}\big\langle\mathsf{ev}\circ((\mathbb{D}\multimap X)\otimes\mathsf{e}_j)\circ\rho^{-1}\circ\mathsf{p}_i\big\rangle_{i\in I}\overset{\circ 2}{=}\langle\pi_j\circ\mathsf{p}_i\rangle_{i\in I} = \&_{i\in I}\,\pi_j\end{aligned}$$

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